Advisor review package -- opx ML thermobarometer

For: Dr. Kanani K.M. Lee **From:** Ta Quang Nhan (cadet, USCGA) **Date:** 2026-04-20 **Target venue:** JGR ML & Computation

What this document is

A single reviewable snapshot of the pyroxene thermobarometer project. Figures come first so you can scan the results visually; data tables come after in case you want to check a specific number.

The project in one paragraph (plain English)

We are trying to estimate the pressure (P) and temperature (T) at which an orthopyroxene (opx) crystal grew, using only the chemistry of that crystal (and optionally its surrounding liquid). Classical petrologic thermobarometers like Putirka (2008) fit a small polynomial to experimental data; we fit a flexible machine-learning model to the same experimental data and compare. The question we are asking is: *does the ML model do better than the polynomial, and if so where and by how much?*

A quick glossary (once, then we move on)

- **opx / cpx**: orthopyroxene and clinopyroxene, two common pyroxene minerals.
- **opx-liq / opx-only**: two prediction pipelines. "opx-liq" uses both the crystal and the surrounding liquid as input; "opx-only" uses just the crystal chemistry.
- **pressure regime**: we pre-registered four geologic pressure bins before any modeling: shallow-crustal (<5 kbar), deep-crustal / MASH (5-15 kbar), lithospheric-mantle (15-30 kbar), deeper-mantle (30+ kbar), plus "ALL" for the combined test set.
- **RMSE**: root-mean-squared error. How far, on average, a prediction misses the truth. Lower is better. Units: C for temperature, kbar for pressure.
- **bias correction**: a small post-processing step applied after the ML model makes its raw prediction. Two flavors are tested: "Form A" learns a regression of residual on predicted value separately in each pressure regime; "Form B" is a single smooth piecewise-sigmoid on the predicted-value axis.
- **ship-if-better rule**: a correction only ships if it improves overall RMSE AND does not make any pre-registered regime worse. A pre-registered promise to not cherry-pick.
- **TabPFN**: a foundation model for small tabular data. In- context transformer architecture, no training, no tuning, no SHAP. We added it as a 9th baseline family.

- **Putirka (2008)**: the reference classical thermobarometer.

How to read this document

1. **Figures first** (Section 1): 13 core figures + 1 SHAP figure. Each figure has a long caption explaining what it shows and what to look for.
2. **Supporting figures** (Section 2): 14 SI figures for the curious.
3. **Data tables** (Section 3): the numbers behind the figures, in case you want to double-check.
4. **Methods, limitations, caveats** (Sections 4-6): short plain-English sections.
5. **Provenance** (Section 7): git state + file inventory.

Sections

1. Core figures (14 PDFs: 13 main + Core_12 SHAP)
2. Supporting information figures
3. Data tables 3.1 Headline (4 opx cells) 3.2 Eight-cell winner table (tuned + TabPFN) 3.3 TabPFN bias-correction scoreboard 3.4 opx-only P per-regime breakdown 3.5 TabPFN head-to-head (pre vs post vs tuned vs Putirka)
4. Methods summary
5. Limitations
6. Caveats and reconstruction provenance
7. Provenance (git SHA, CSV inventory)
8. Pre-registration (verbatim)

In [1]:

```
from pathlib import Path
import json
import pandas as pd
from IPython.display import display, Markdown, Image

ROOT = Path('.').resolve().parent.parent  # deliverables/lee_package_20260420/ -> p
RESULTS = ROOT / 'results'
FIGS = Path('./figures')

def load_csv(relpath: str) -> pd.DataFrame:
    p = ROOT / relpath
    if not p.exists():
        raise FileNotFoundError(f'required CSV missing: {p}')
    return pd.read_csv(p)

def load_json(relpath: str) -> dict:
    p = ROOT / relpath
    if not p.exists():
        raise FileNotFoundError(f'required JSON missing: {p}')
    return json.loads(p.read_text())
```

```
pd.set_option('display.float_format', lambda x: f'{x:.3f}')
print('Setup complete; loader ready.')
```

```
Setup complete; loader ready.
```

1. Core figures

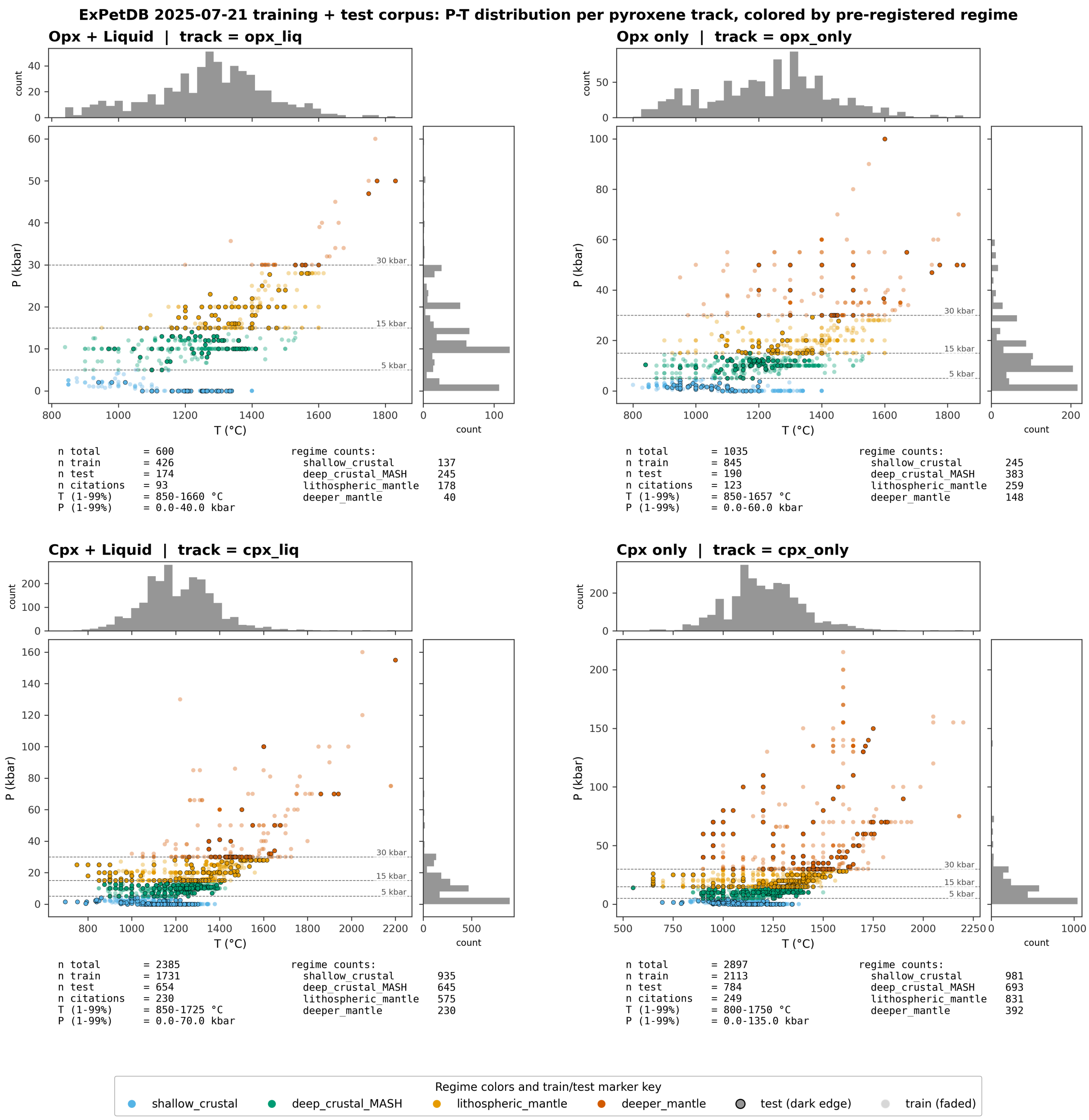
Fourteen figures in total. Each one is paired with a long caption written in plain language. You don't need to load any data to read this section.

- **Core_01 / Core_01b**: who is in the training set, who is in the held-out test set (the sanity-check that no single citation leaks across the split).
- **Core_02**: the citation-grouped fold split used for cross- validation.
- **Core_03**: a one-page flowchart of the ML pipeline.
- **Core_04**: cross-pipeline heatmap of model RMSE.
- **Core_05**: per-regime RMSE before and after bias correction.
- **Core_06**: residual-vs-prediction scatter at canonical seed 42.
- **Core_07**: scoreboard of ML vs Putirka, regime by regime.
- **Core_08**: the headline per-regime RMSE bar chart for all four opx cells.
- **Core_09a / Core_09b**: how each ML family compares, by regime and overall.
- **Core_10**: best ML model vs Putirka, one dot per test sample.
- **Core_11**: natural-sample two-pyroxene 1:1 agreement plot.
- **Core_12 (new)**: SHAP feature importance for the best- explainable model per cell. SHAP tells you which oxide features the model was actually leaning on.

Core_01_fig_dataset_map

Figure 1. Dataset map. P-T distribution of all experiments in the ExPetDB 2025-07-21 export, partitioned into four pyroxene tracks (opx-liq, opx-only, cpx-liq, cpx-only). Points are colored by pre-registered pressure regime (shallow_crustal <5 kbar, deep_crustal_MASH 5-15 kbar, lithospheric_mantle 15-30 kbar, deeper_mantle >=30 kbar); regime edges were locked on 2026-04-17 before any correction fitting. Dashed horizontal lines mark regime boundaries (labeled at right axis). Marker opacity distinguishes the 80/20 citation-grouped train/test split: train points faded, test points with dark edges. Marginal histograms show the univariate T and P coverage. Each panel is titled with its track name. Directly below each scatter, a per-panel two-column summary reports n_total, n_train, n_test, citation count, and 1st-99th percentile T and P ranges on the left, with per-regime counts on the right. The regime/split color key is shown as a single shared legend at the bottom.

In [2]:
```
from IPython.display import Image
png = FIGS / 'Core_01_fig_dataset_map.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

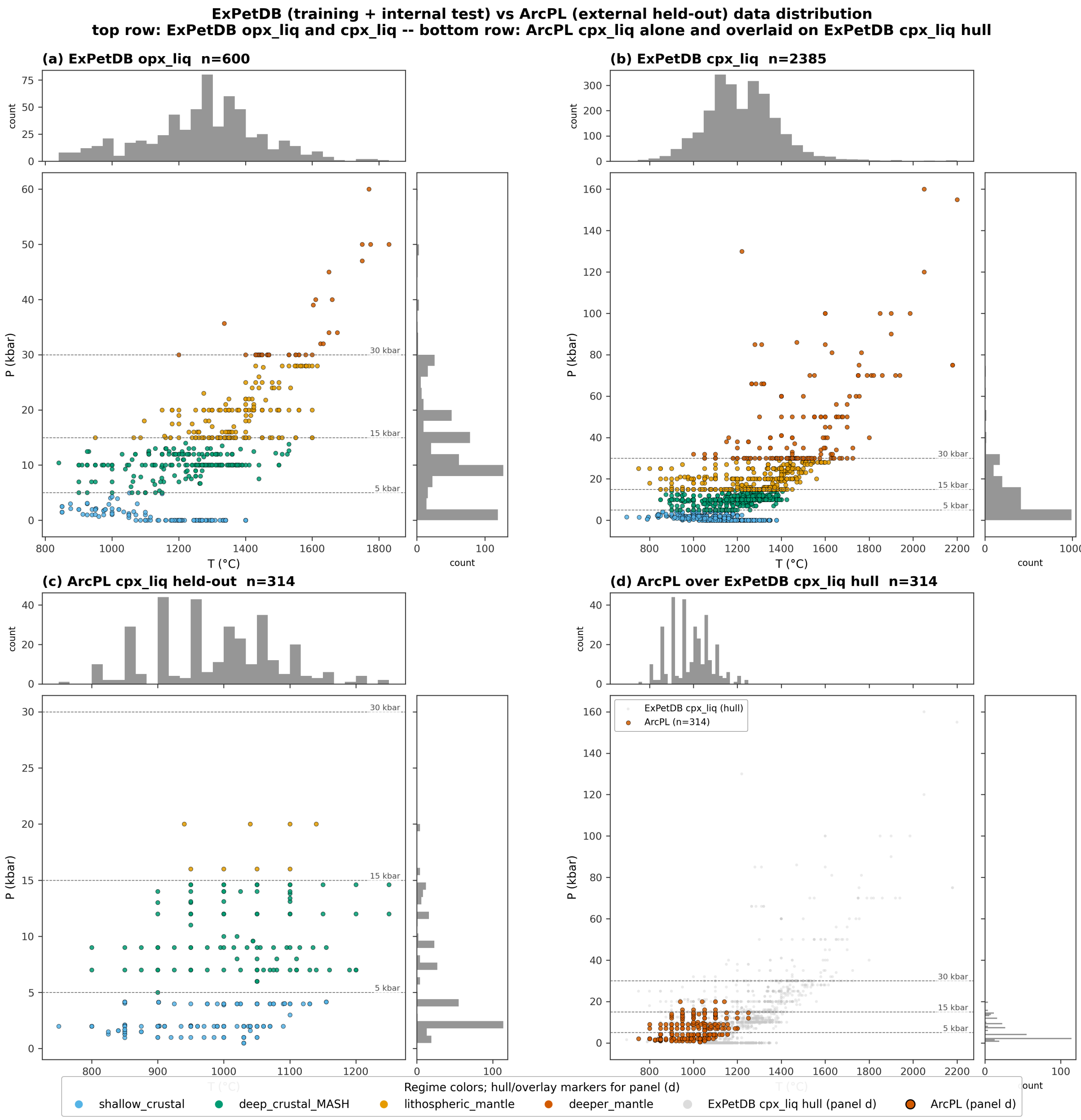


Core_01b_fig_dataset_map_holdout

Figure 1b. Distribution comparison between the ExPetDB training corpus and the ArcPL external held-out dataset. Top row: ExPetDB opx_liq (panel a) and cpx_liq (panel b) P-T scatter, colored by pre-registered pressure regime with marginal T and P histograms. Bottom row: ArcPL cpx_liq held-out (panel c, regime colored) and ArcPL overlaid on the ExPetDB cpx_liq hull (panel d, gray shows ExPetDB extent, vermillion shows ArcPL). ArcPL is cpx- only (n=314 filtered to finite P/T) and serves as an out-of- distribution check for the cpx pipeline. Regime boundary dashed lines are labeled at the right edge of each plot using axis-fraction coordinates so labels always stay inside the axes. Source: data/processed/{opx_clean_opx_liq, cpx_clean_cpx_liq}.parquet and data/external/agreda_lopez_2024/.../ArcPl_filtered.xlsx.

In [3]:
```
from IPython.display import Image
png = FIGS / 'Core_01b_fig_dataset_map_holdout.png'
```

```
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

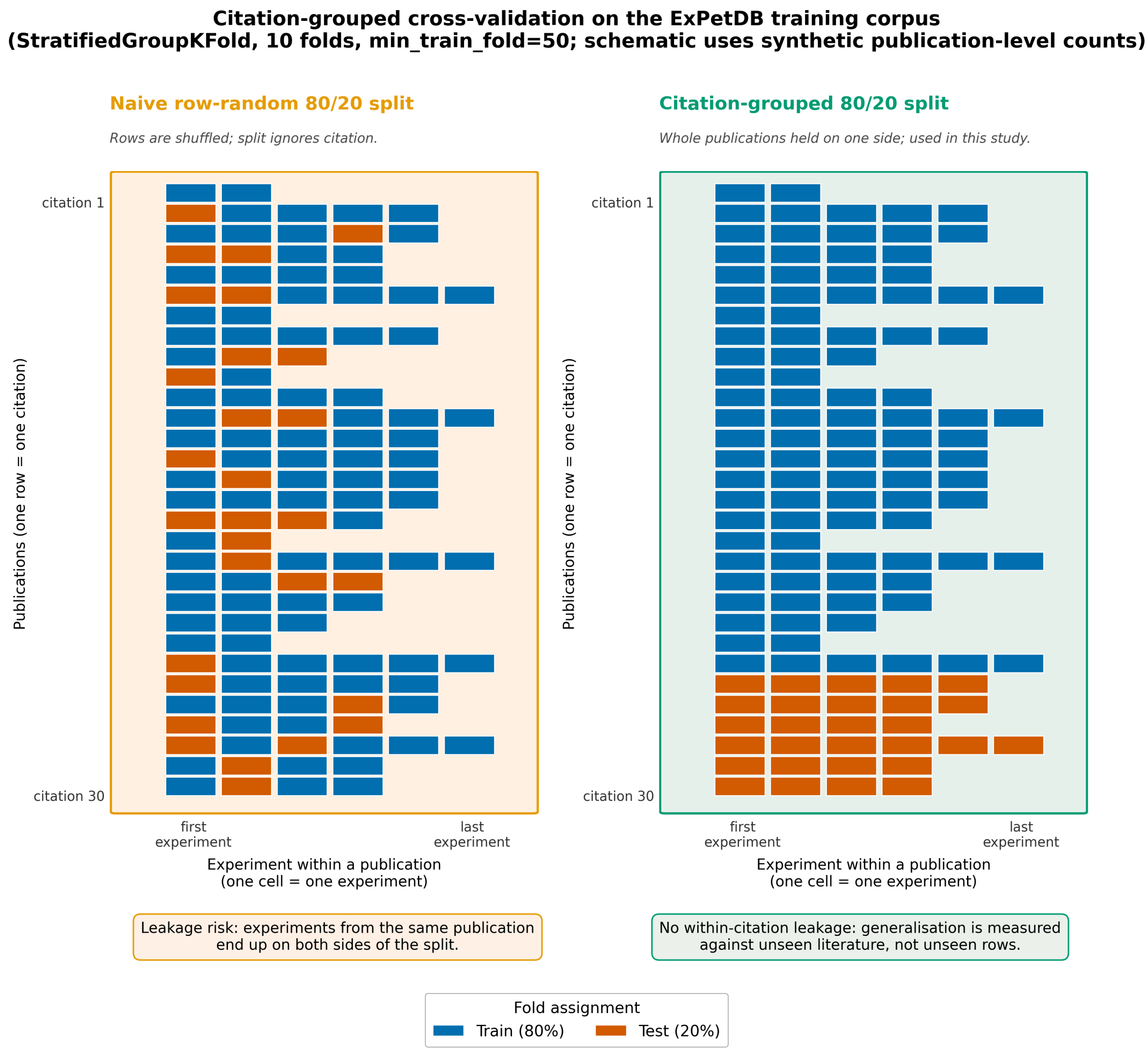


Core_02_fig_citation_split

Figure 2. Citation-grouped cross-validation schematic on the ExPetDB training corpus (opx n=1635, cpx n=5282 after equilibrium and cation filters). (a) A naive row-random 80/20 split lets experiments from the same publication appear on both sides of the split; correlated features (same laboratory, same protocol, same starting materials) leak from train to test and inflate generalisation estimates. (b) The citation-grouped split used throughout this study: every publication is held wholly on one side. We use scikit-learn StratifiedGroupKFold with 10 folds and min_train_fold=50 to stratify on pressure regime while respecting citation

grouping. Each cell represents one experiment; each row is one publication (~93 publications in the opx corpus, ~260 in the cpx corpus).

```
In [4]: from IPython.display import Image
png = FIGS / 'Core_02_fig_citation_split.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

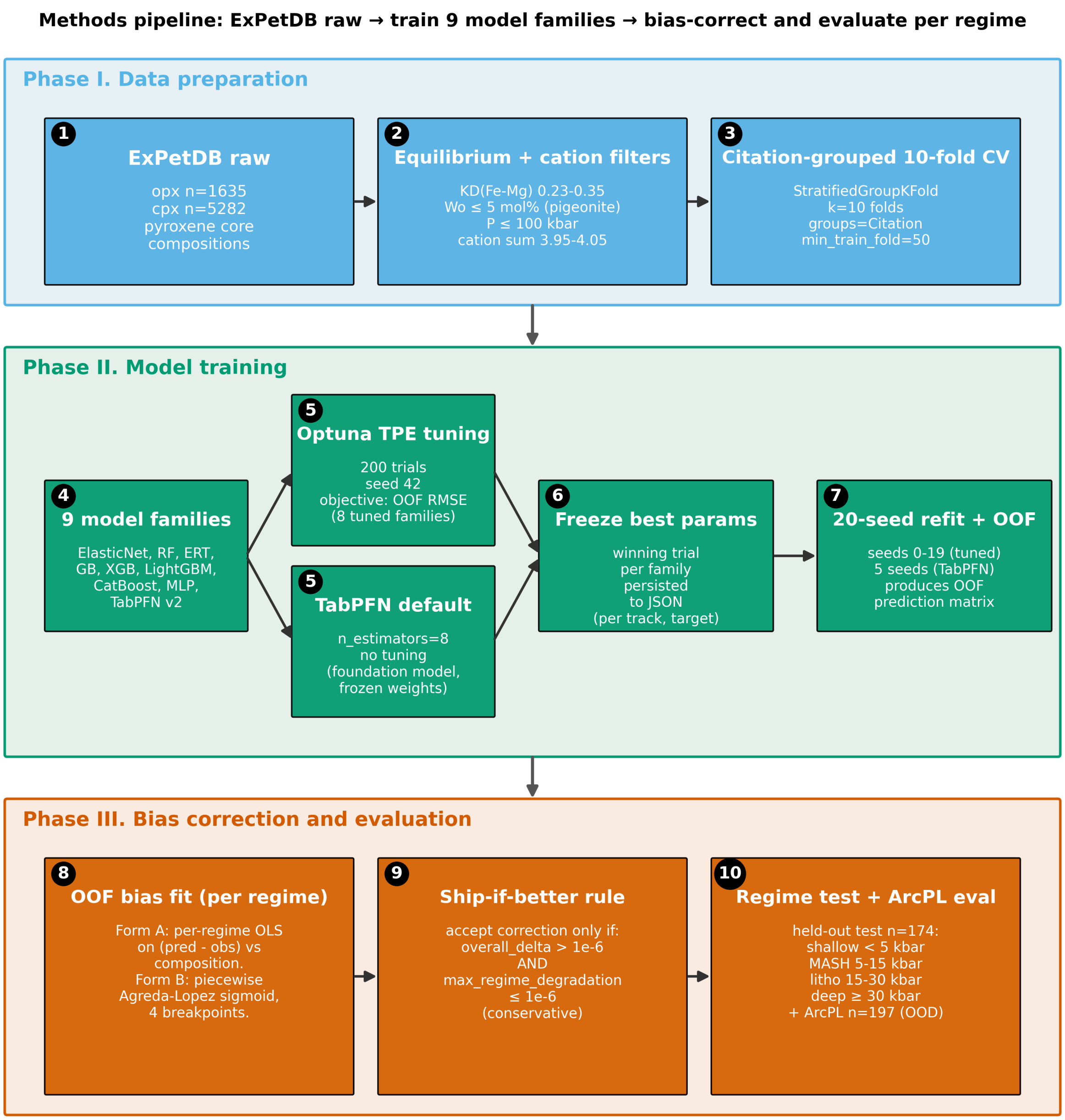


Core_03_fig_methods_flowchart

Figure 3. Methods flowchart. Three phase panels, each containing numbered stages that flow strictly left-to-right. Phase I (Data): raw ExPetDB (stage 1), filtering (stage 2), citation-grouped 10-fold split using StratifiedGroupKFold with min_train_fold=50 (stage 3). Phase II (Model training): 9 model families (stage 4) split into two tuning routes at stage 5 (Optuna 200 trials, seed 42 for the eight tuned families, vs. TabPFN default at n_estimators=8), followed by parameter freeze (stage 6) and a 20-seed refit (5-seed for TabPFN to bound CPU cost) producing out-of-fold predictions (stage 7). Phase III (Evaluation): OOF bias fit for Form A

(per-regime OLS) and Form B (piecewise Agreda-Lopez sigmoid) (stage 8); conservative ship-if-better rule with overall_delta > 1e-6 AND max_regime_degradation <= 1e-6 (stage 9); evaluation on two slices at stage 10 -- the held-out ExPetDB test partition split into the four pre-registered pressure regimes, plus the ArcPL n=197 external-validation dataset as an out-of-distribution check.

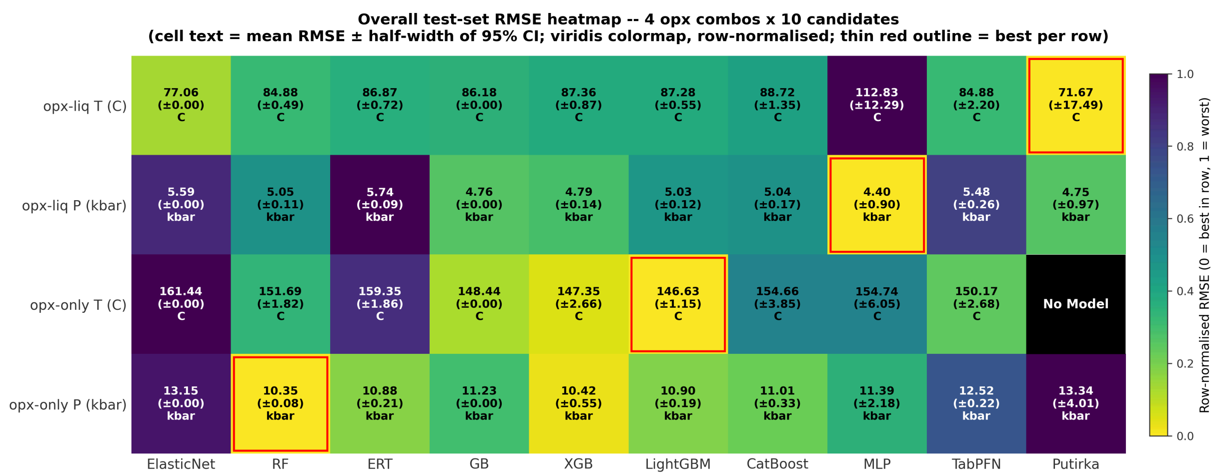
In [5]:
```python
from IPython.display import Image
png = FIGS / 'Core_03_fig_methods_flowchart.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```



Core_04_fig_nb04_cross_pipeline_heatmap

Figure 4. Overall test-set RMSE heatmap across the four opx track/target combinations (rows) versus ten candidates (columns: the nine ML model families in canonical MODEL_ORDER plus the best Putirka / Agreda external benchmark). Each cell reports the mean RMSE of the 20-seed refit (5 seeds for TabPFN) over the held-out test partition, together with the half-width of the 95% confidence interval (1.96 * std for the ML families, half the bootstrap CI for Putirka). Cells are colored via a row-normalised viridis_r colormap so the row minimum (best) is lightest and the row maximum (worst) is darkest; absolute values are printed in each cell in native units (C for T targets, kbar for P targets). Opx-only T has no Putirka opx-only thermometer available in Thermobar and is marked N/A. Per-family feature_set choice (raw / alr / pwlr) is the 20-seed best per row. The minimum-RMSE cell in each row is outlined in thin red to highlight the best model per opx combo.

In [6]:
```python
from IPython.display import Image
png = FIGS / 'Core_04_fig_nb04_cross_pipeline_heatmap.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

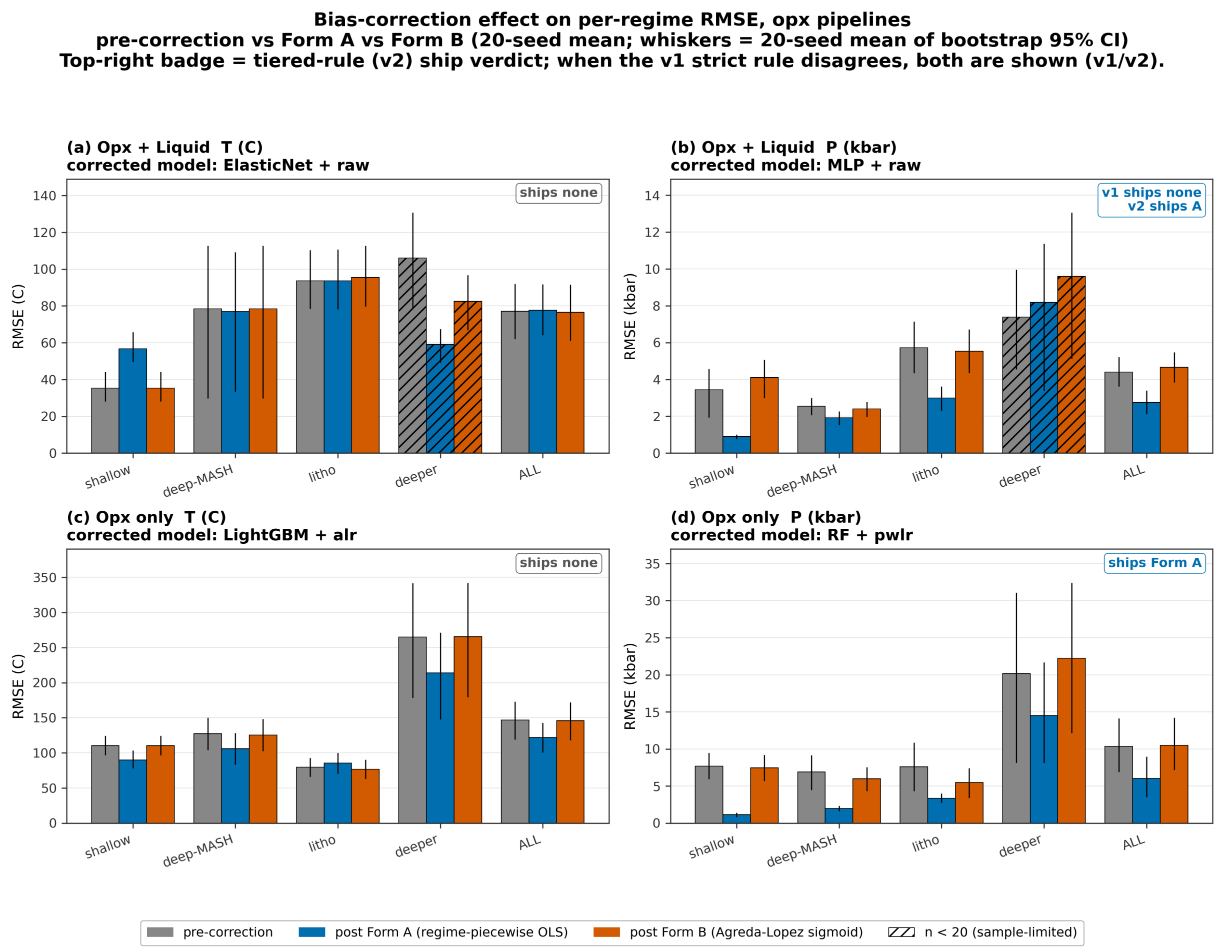


Core_05_fig30_bias_correction_per_regime_rmse

Figure 5. Bias-correction effect on per-regime test-set RMSE for the four opx track/target combinations. Each regime group shows three bars: pre-correction (gray), post Form A (regime-piecewise OLS, blue), post Form B (Agreda-Lopez piecewise sigmoid, vermillion). Bar heights are 20-seed means; whiskers are the 20-seed mean of per-seed bootstrap 95% CIs. Hatched bars mark regimes with n < 20 (sample-size limited, interpret with care). The shipped correction form per panel (accepts Form A / accepts Form B / ships none) is annotated in the top-right; a form only ships when it improves overall RMSE and does not degrade any regime by more than 1e-6. How the math works. Both forms post-process the raw ML prediction y_hat using the out-of-fold residual e = y_hat - y learned on the training partition, then subtract the learned residual at test time (y_corr = y_hat - e_hat). Form A (regime-piecewise OLS) fits a separate linear regression e = a + b * y_hat on each pre-registered P-regime, so the correction is a constant slope/intercept within each regime (four

regressions stitched end-to-end by regime boundary). Form B (Agreda-Lopez sigmoid) fits one smooth piecewise-sigmoid curve e(y_hat) across all regimes with four learned breakpoints (following Agreda-Lopez et al., 2024), which avoids the discontinuities at regime edges that Form A introduces but adds more degrees of freedom. A form "ships" only if it reduces overall RMSE and does not hurt any regime. Cpx pipelines are intentionally excluded; this paper centers the opx ML thermobarometer. Source CSVs: results/bias_correction_per_seed.csv, results/bias_correction_shipped.csv.

In [7]:
```
from IPython.display import Image
png = FIGS / 'Core_05_fig30_bias_correction_per_regime_rmse.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

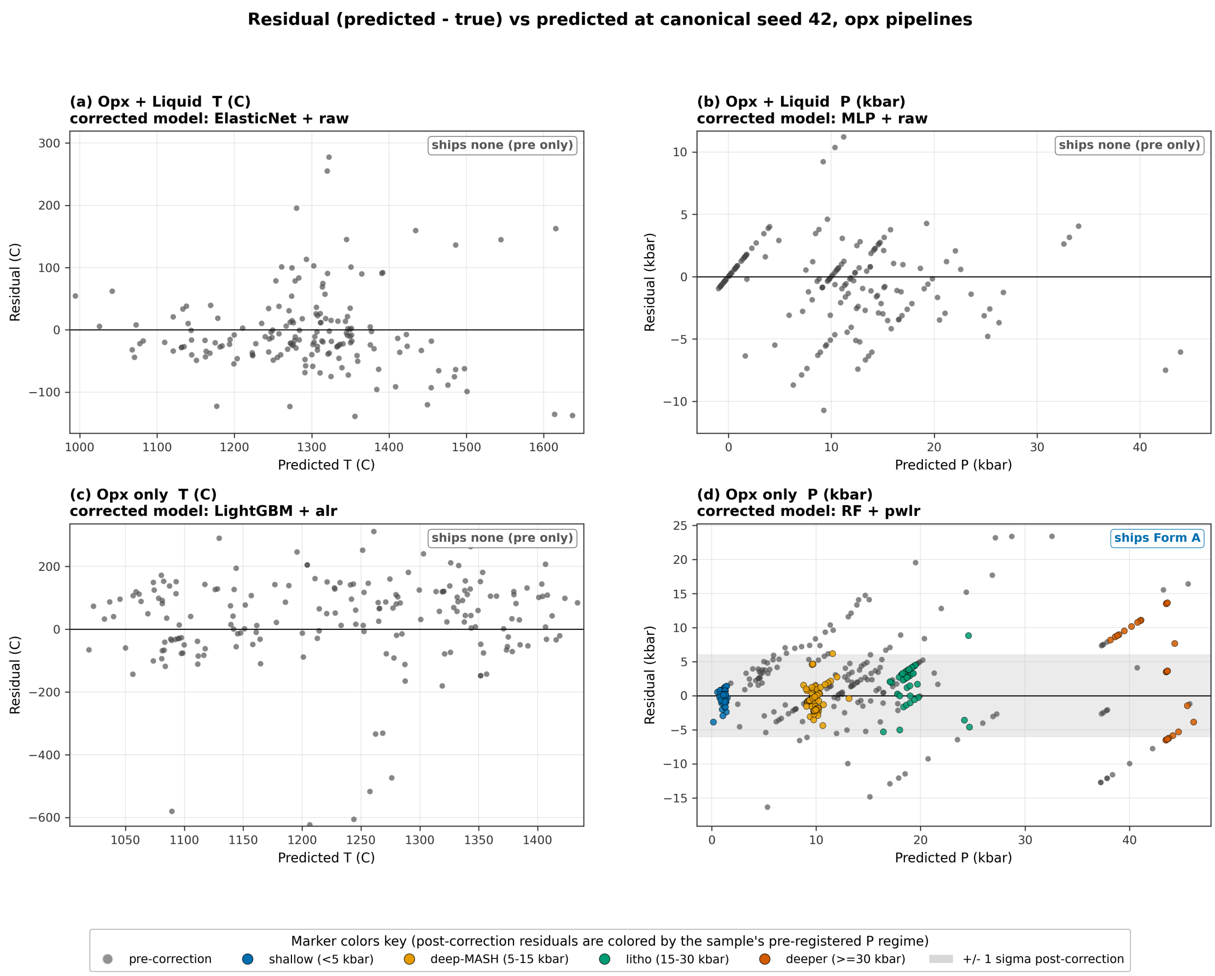


Core_06_fig31_bias_correction_residuals

Figure 6. Residual-vs-predicted scatter at canonical seed 42 for the four opx track/target combinations. Gray markers: pre-correction residuals. Colored markers: residuals after the shipped form of bias correction, colored by the sample's pre-registered P regime (shallow_crustal blue, deep_crustal_MASH orange, lithospheric_mantle green, deeper_mantle vermillion). A gray band marks +/-1 sigma of the post-correction residuals. Panels whose

shipping decision is "none" show the pre-correction residuals alone and are labelled accordingly. The figure-level legend describes marker colors and applies to every panel, including the "ships none" panels where the colored markers would otherwise not appear. Source: results/bias_correction_shipped.csv and per-sample predictions from results/bias_correction/checkpoints/ (seed 42).

In [8]:
```
from IPython.display import Image
png = FIGS / 'Core_06_fig31_bias_correction_residuals.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

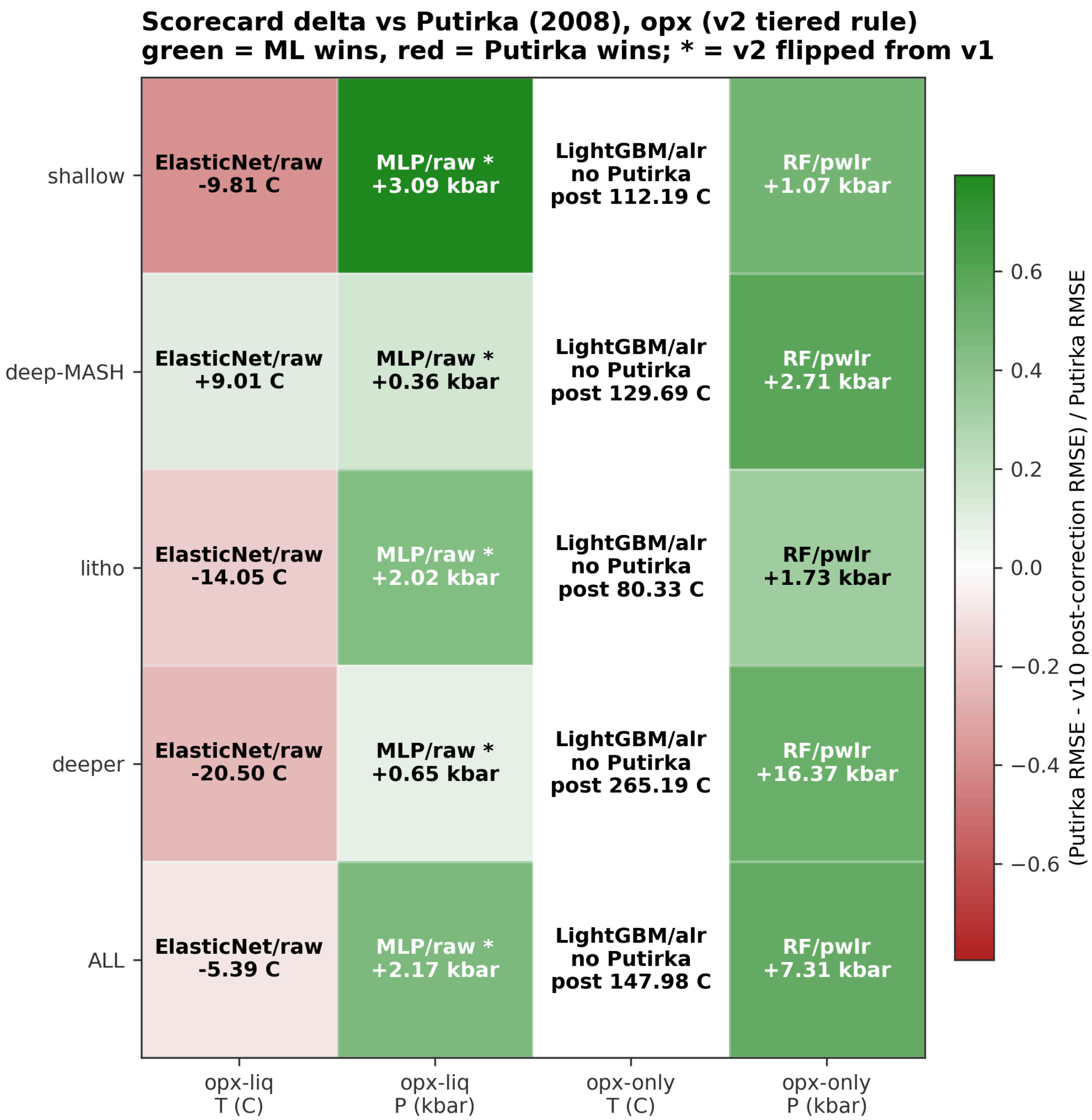


Core_07_fig34_bias_correction_scorecard_delta

Figure 7. Per-regime scorecard delta between our post-correction-bias-correction RMSE and the best external Putirka (2008) thermobarometer, for the four opx track/target cells. Color encodes fractional improvement (Putirka_RMSE - v10_post_RMSE) / Putirka_RMSE on a diverging red-white-green scale: green cells are wins for the tuned ML baseline, red cells are wins for Putirka. Cell text shows the absolute RMSE delta in native units (C for T, kbar for P). External references are Putirka (2008) thermobarometers run through Thermobar: opx-liq T = eq 28a, opx-liq P = eq 29a (29b in deeper_mantle), opx-only P = eq 29c. Opx-only T has

no Putirka opx-only thermometer in Thermobar, so its row reports the tuned ML baseline absolute RMSE instead of a delta. Cells marked with "*" had their post-correction RMSE flip under the v2 tiered ship rule (Amendment 1, 2026-04-20) because a low-n regime that had vetoed the correction under v1 is non-vetoing under v2. Cpx pipelines are intentionally excluded. Source: results/preregistered_scorecard_postcorrection_v2.csv (with v1 reference from preregistered_scorecard_postcorrection.csv).

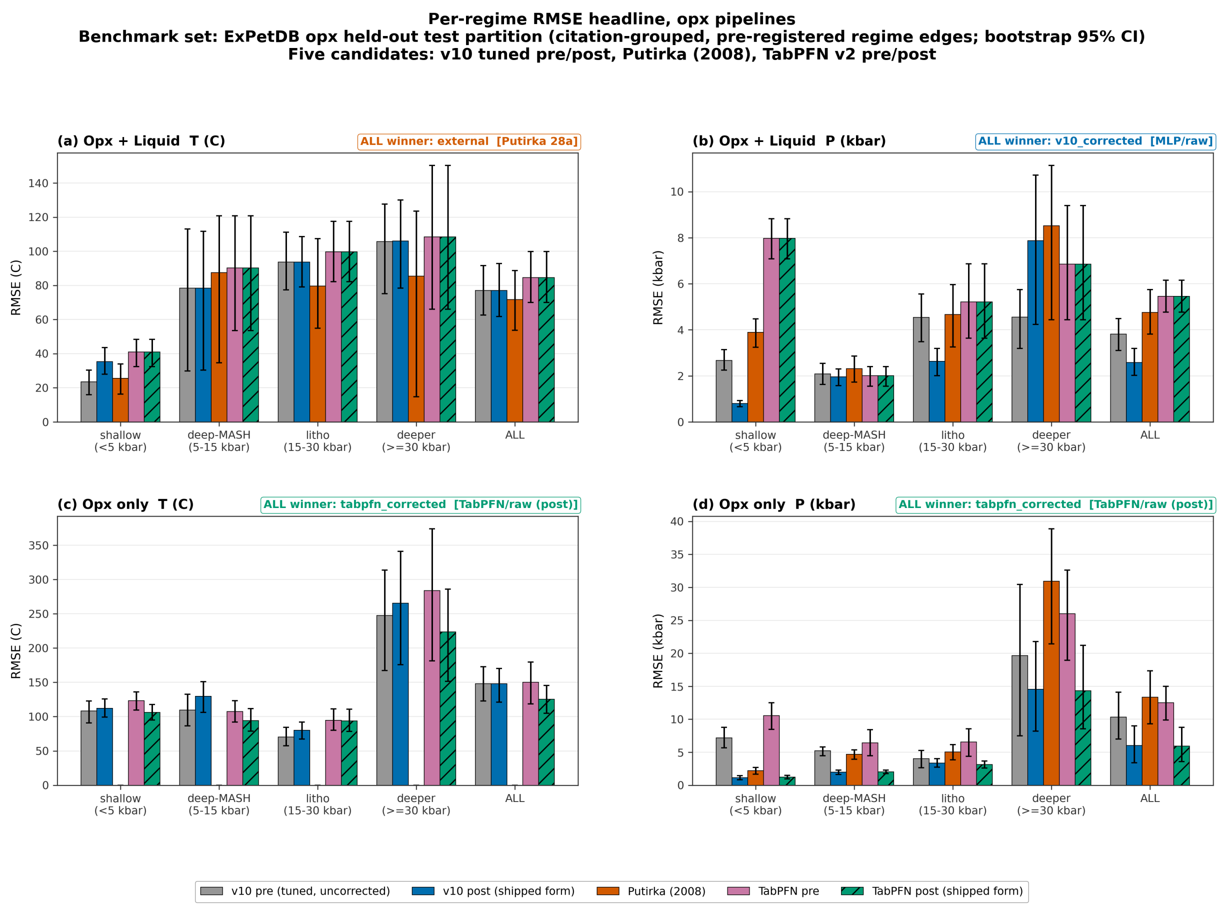
In [9]:
```
from IPython.display import Image
png = FIGS / 'Core_07_fig34_bias_correction_scorecard_delta.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```



Core_08_fig45_opx_headline

Figure 8. Per-regime RMSE for all four opx track/target combinations across five candidates: pre-correction tuned baseline-correction (gray), post-correction tuned baseline-correction in its shipped form (blue), best external Putirka (2008) thermobarometer (red), TabPFN v2 pre-correction (pink), TabPFN v2 post-correction in its shipped form (green, hatched). Regimes follow the pre-registered pressure partition (<5, 5-15, 15-30, >=30 kbar) plus ALL. Error bars are 95% bootstrap CIs (n_boot = 500). "ALL winner" in each panel is the preregistered scorecard verdict, enriched with the winning model family / feature-set (e.g. ERT/pwlr = ERT trained on pairwise-log-ratio transformed oxide features). Opx-only T (panel c) has no Putirka opx-only thermometer in Thermobar, so the red bar is absent. Source: results/preregistered_scorecard_postcorrection.csv.

In [10]:
```python
from IPython.display import Image
png = FIGS / 'Core_08_fig45_opx_headline.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

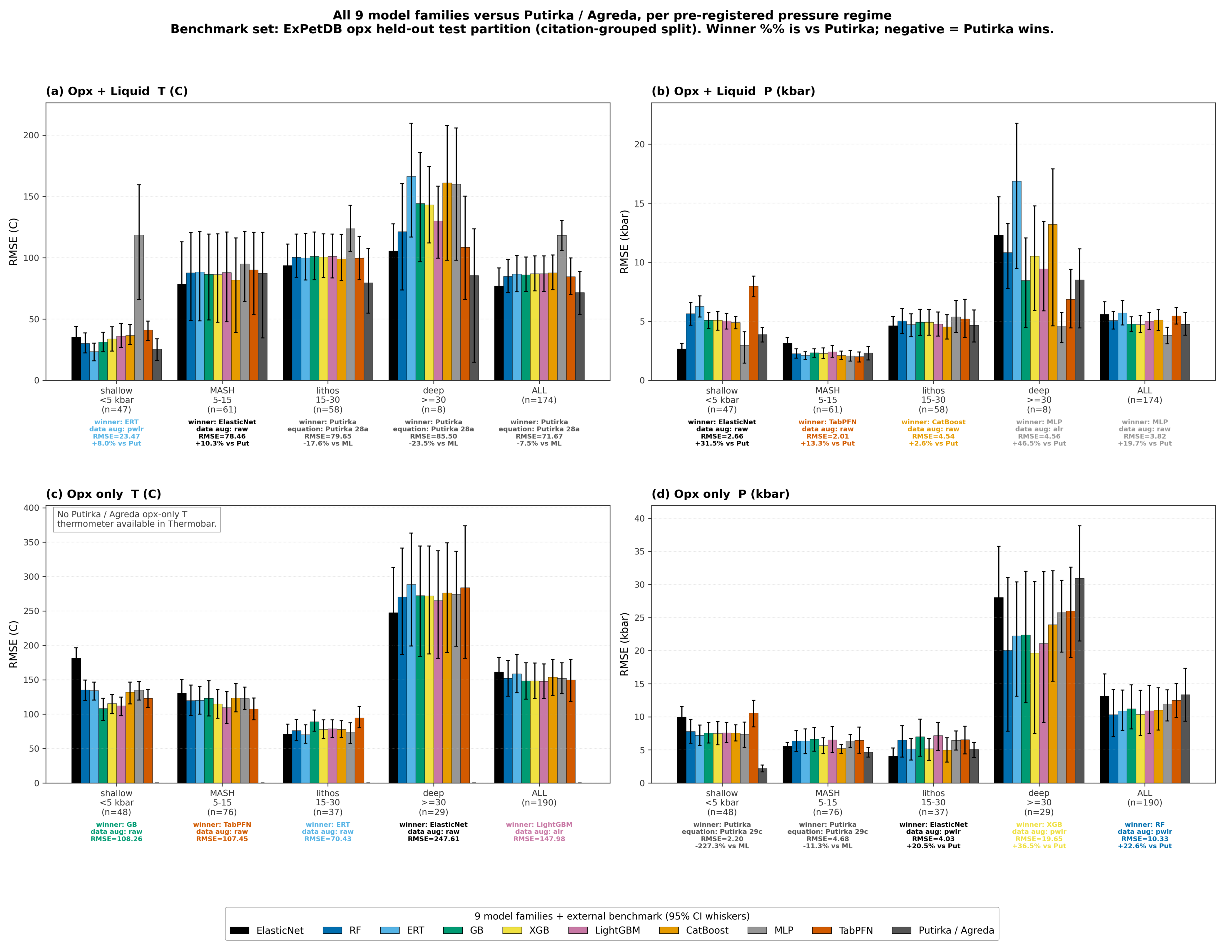


Core_09a_fig_opx_regime_families

Figure 9a. Per-regime test-set RMSE for all nine model families versus the best Putirka / Agreda external benchmark, across the four opx combinations. Each regime group contains ten bars: nine colored bars (one per family, in the canonical MODEL_ORDER) plus a gray

Putirka / Agreda bar. For each (family, regime) cell the feature_set (raw / alr / pwlr) with the lowest test RMSE is selected; whiskers are 95% bootstrap CIs. Pressure regimes are the pre-registered edges shallow_crustal <5 kbar, deep_crustal_MASH 5-15 kbar, lithospheric_mantle 15-30 kbar, deeper_mantle >=30 kbar, plus the ALL aggregate column. Panel (c) opx-only T has no Putirka opx-only thermometer in Thermobar and is annotated accordingly. Core_09b is the overall-only companion; Core_10 collapses to best-of-ours vs Putirka.

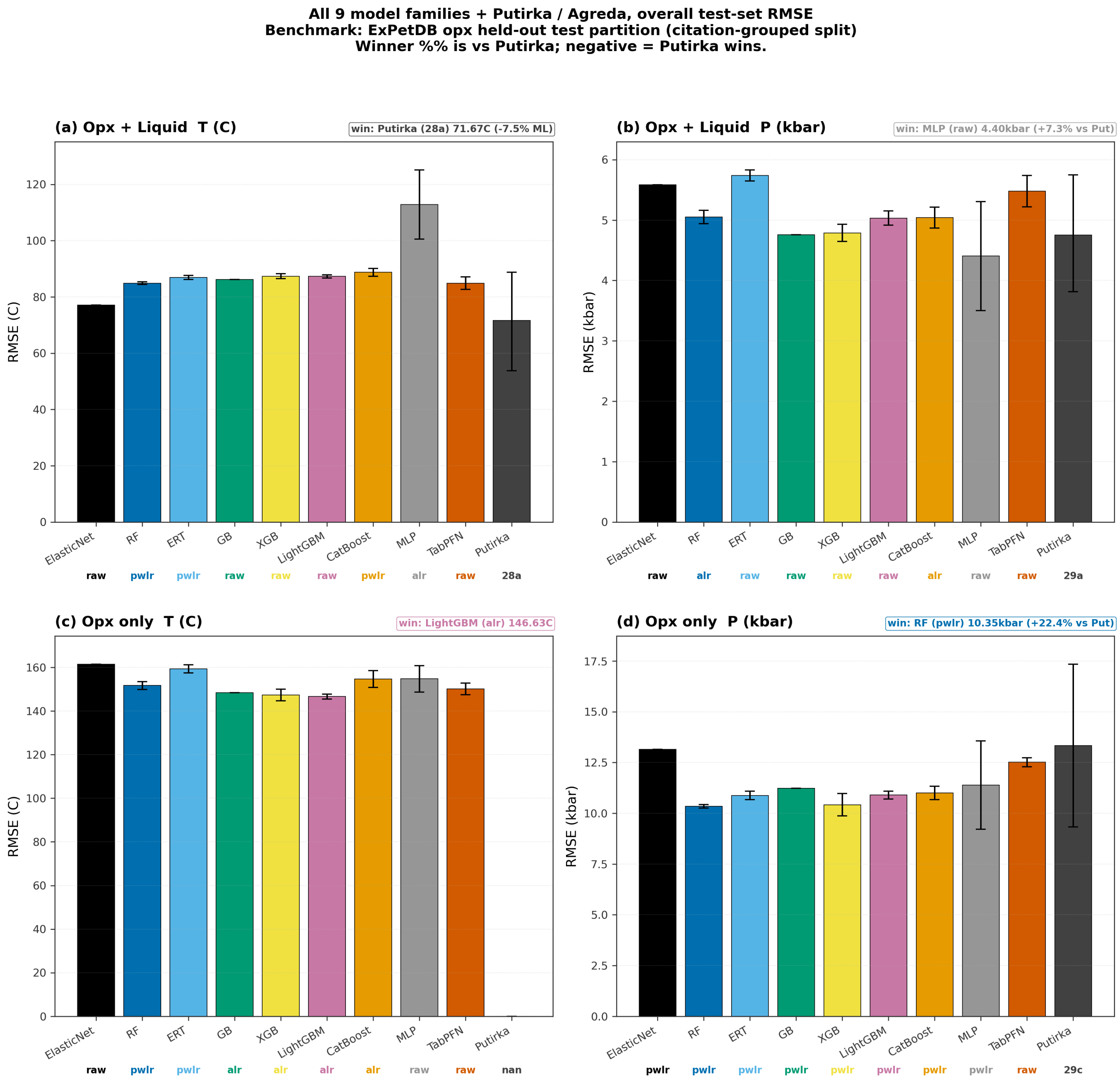
In [11]:
```python
from IPython.display import Image
png = FIGS / 'Core_09a_fig_opx_regime_families.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```



Core_09b_fig_opx_overall_families

Figure 9b. Overall test-set RMSE per model family for all four opx combinations, aggregated across the full held-out partition (no regime split). Bars show the mean 20-seed RMSE (5-seed for TabPFN), whiskers are 95% intervals computed as mean +/- 1.96 * std over seeds; the feature-set winner (raw, alr, or pwlr) for each family is printed below the bar in the family color. The dashed horizontal line is the best Putirka / Agreda external benchmark from the ALL-row of the pre-registered scorecard (shaded band = 95% bootstrap CI on the benchmark); panel (c) opx-only T has no Putirka opx-only thermometer in Thermobar and is annotated accordingly. This figure answers "which family is best overall?"; the companion Core_09a answers the same question stratified by pressure regime.

In [12]:
```python
from IPython.display import Image
png = FIGS / 'Core_09b_fig_opx_overall_families.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

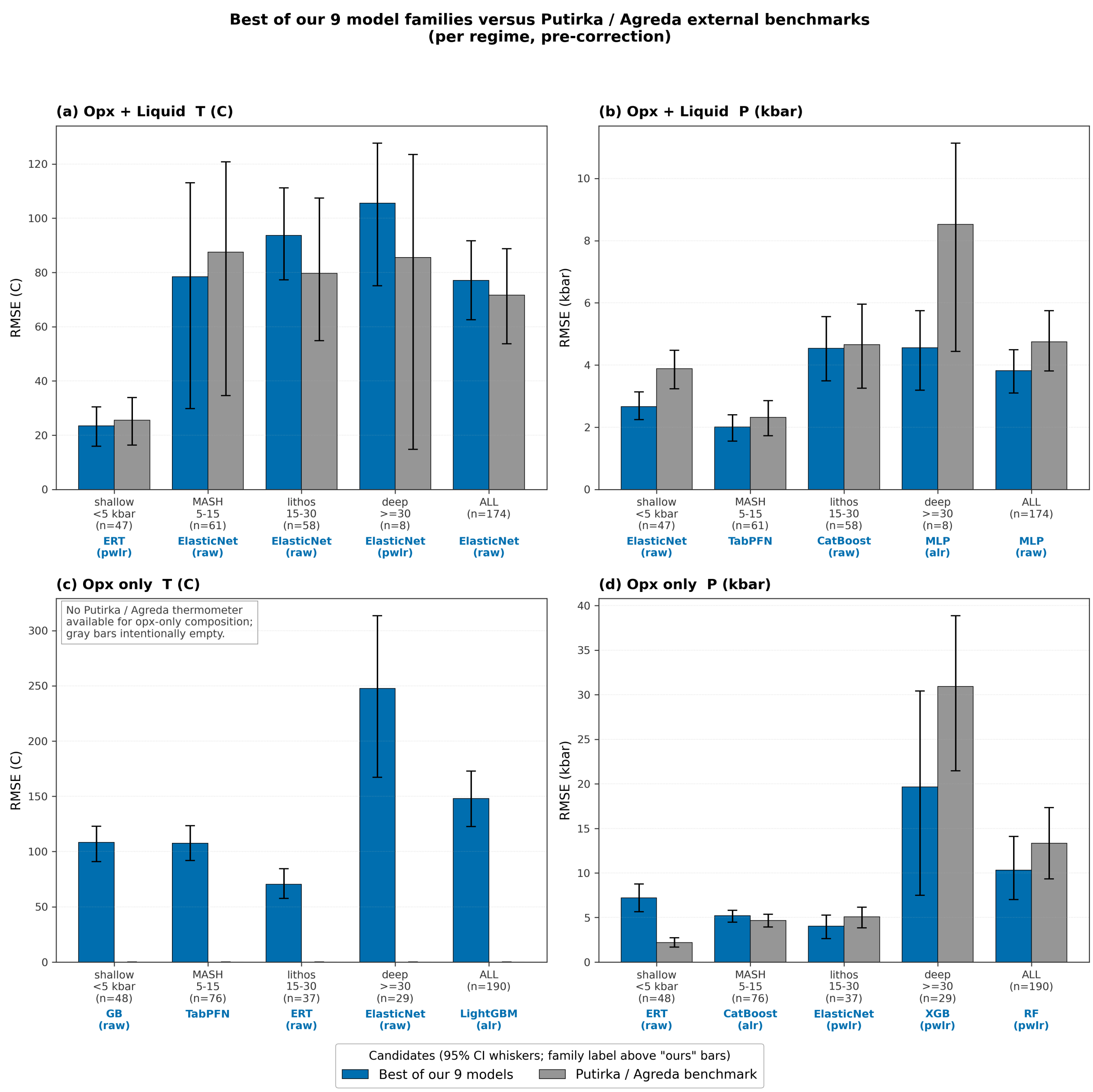


Core_10_fig_best_vs_putirka

Figure 10. Best of our nine model families versus Putirka / Agreda external benchmarks, per pre-registered pressure regime and aggregate ALL column, for all four opx combinations. For each (track, target, regime) the "ours" bar is the minimum pre-correction RMSE across ElasticNet, RF, ERT, GB, XGB, LightGBM, CatBoost, MLP, and TabPFN; the family whose model

won that regime is labelled below its bar together with the input transform ("raw" = native oxide wt%, "alr" = additive log-ratio, "pwlr" = pairwise log-ratio) that won for that family / regime (example: ERT + pwlr). The "Putirka" bar is the best external Thermobar/Agreda benchmark from the head-to-head comparison. Whiskers are 95% bootstrap confidence intervals from the 20-seed refit (5 seeds for TabPFN). Panel (c) opx-only T has no Putirka opx-only thermometer available in Thermobar and is annotated accordingly. This is a simplified companion to Core_09 (all 9 families shown individually); Core_10 answers the narrower question "does our best ML candidate beat the best published calibration at each regime?" at a glance.

In [13]:
```python
from IPython.display import Image
png = FIGS / 'Core_10_fig_best_vs_putirka.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

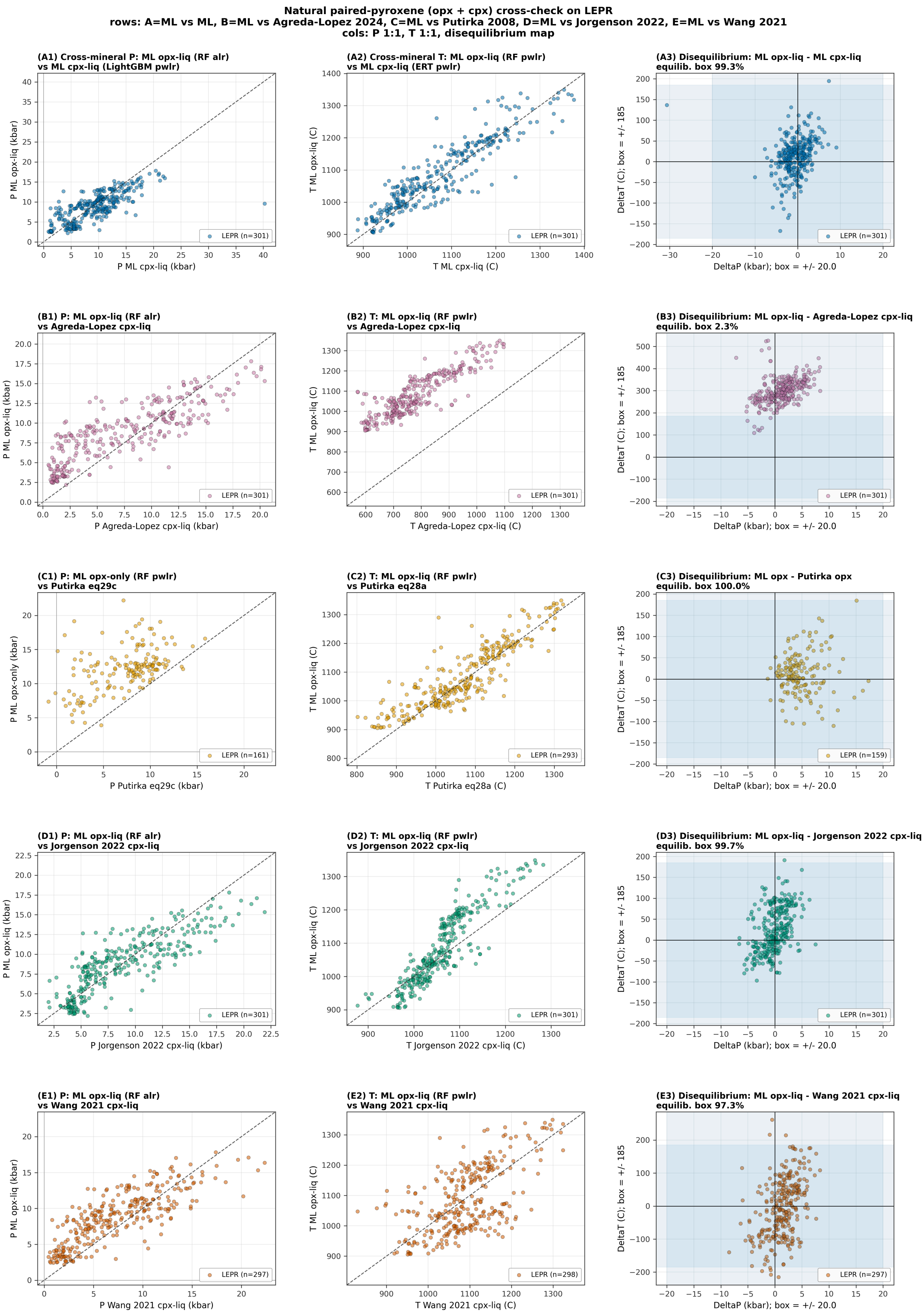


Core_11_fig_nb08_twopx_1to1

Figure 11. Natural paired-pyroxene sanity check on LEPR samples that contain both opx and cpx (inner-join on Experiment). Five row categories, each presented as (P 1:1, T 1:1, disequilibrium map). Row A compares our ML opx-liq (RF pwlr T, RF alr P) with our ML cpx-liq (ERT pwlr T, LightGBM pwlr P) -- cross-mineral agreement between two independently trained ML models. Row B compares our ML opx-liq with the external Agreda-Lopez (2024) cpx-liq model. Row C compares our ML opx predictions with the classical Putirka (2008) opx thermobarometers (opx-only P via eq 29c, opx-liq T via eq 28a; same-mineral benchmark). Row D compares our ML opx-liq with the Jorgenson (2022) ET-based cpx-liq model. Row E compares our ML opx-liq with the Wang (2021) XGB-based cpx-liq model. Disequilibrium maps plot DeltaP vs DeltaT; the shaded box spans +/- 2 sigma_conformal (from nb07 calibration, used as a petrological equilibrium flag). Axes are auto-scaled so no samples are clipped. Source data: LEPR_Wet_Stitched (April 2023 dump); intermediate predictions cached in results/core11_extended_predictions.csv.

In [14]:

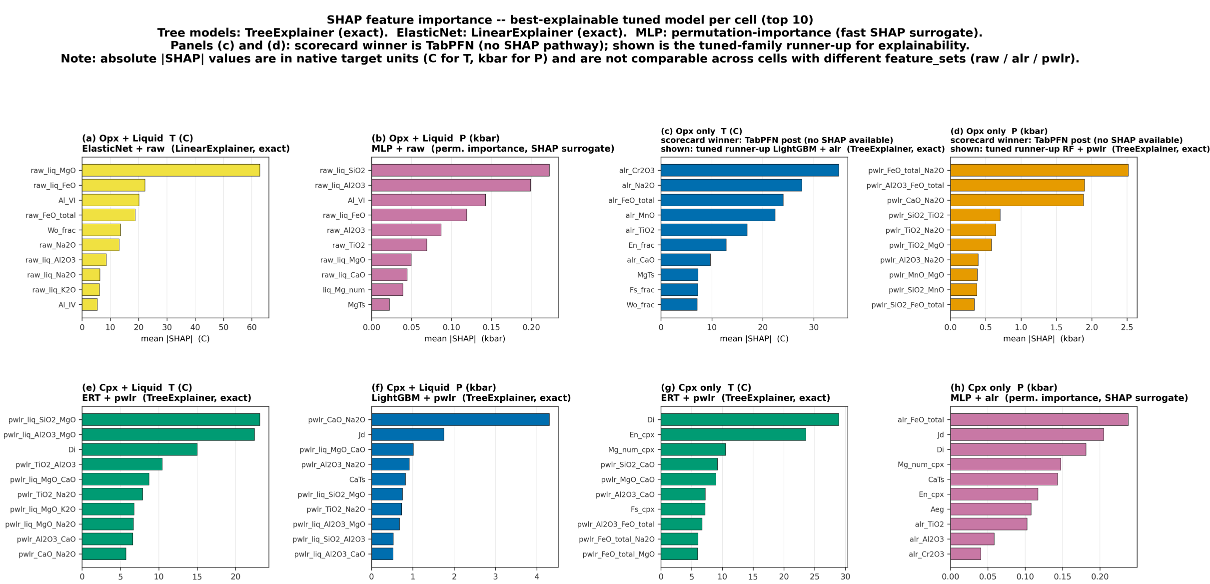
```
from IPython.display import Image
png = FIGS / 'Core_11_fig_nb08_twopx_1to1.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```



Core_12_fig_shap_winners

Figure 12. SHAP feature importance for the best-explainable tuned-family model in each of the eight pipeline x track x target cells. Each panel shows the top 10 features ranked by mean |SHAP| on the held-out test partition at canonical seed 42. Bar length is mean |SHAP| in the target's native units (C for temperature, kbar for pressure). Tuned-family identities: opx_liq T = ElasticNet/raw, opx_liq P = MLP/raw, opx_only T = LightGBM/alr, opx_only P = RF/pwlr, cpx_liq T = ERT/pwlr, cpx_liq P = LightGBM/pwlr, cpx_only T = ERT/pwlr, cpx_only P = MLP/alr. IMPORTANT: for opx_only T (panel c) and opx_only P (panel d) the 5-way scorecard (results/preregistered_scorecard_postcorrection.csv, ALL regime) promotes TabPFN post-correction over the tuned-family winner (TabPFN post-RMSE 125.6 C vs LightGBM/alr 148.0 C, and 5.94 kbar vs RF/pwlr 6.03 kbar respectively). TabPFN is an in-context foundation model and exposes no SHAP pathway (no TreeExplainer, no LinearExplainer, no gradient surface compatible with KernelExplainer at our runtime budget), so those two panels show the tuned-family runner-up's SHAP for explainability. Tree-family (RF, ERT, LightGBM) uses shap.TreeExplainer (exact); ElasticNet uses shap.LinearExplainer (exact) on scaler-transformed features; MLP uses sklearn.inspection.permutation_importance (20 repeats, seed=42) as a fast SHAP surrogate. KernelExplainer for MLP would add ~30 min per cell and was deferred in this pass (Phase 6 one-shot scope, 2026-04-20). Absolute |SHAP| values are NOT comparable across cells with different feature_sets (raw, alr, pwlr) because those transforms rescale the input. Source: results/shap_importance_winners.csv; scorecard winners from results/preregistered_scorecard_postcorrection.csv.

In [15]:
```python
from IPython.display import Image
png = FIGS / 'Core_12_fig_shap_winners.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```



2. Supporting information figures

Extra figures for the curious reader. Safe to skim.

fig24_per_regime_rmse_opx_liq

Figure 24. Per-regime RMSE (bootstrap 95% CI) for the opx-liq test split: best tuned model cell vs Putirka 2008 best equation, in the pre-registered four-bin P partition. Shaded bins are sample-size-limited (n<20).

In [16]:
```
from IPython.display import Image
png = FIGS / 'fig24_per_regime_rmse_opx_liq.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

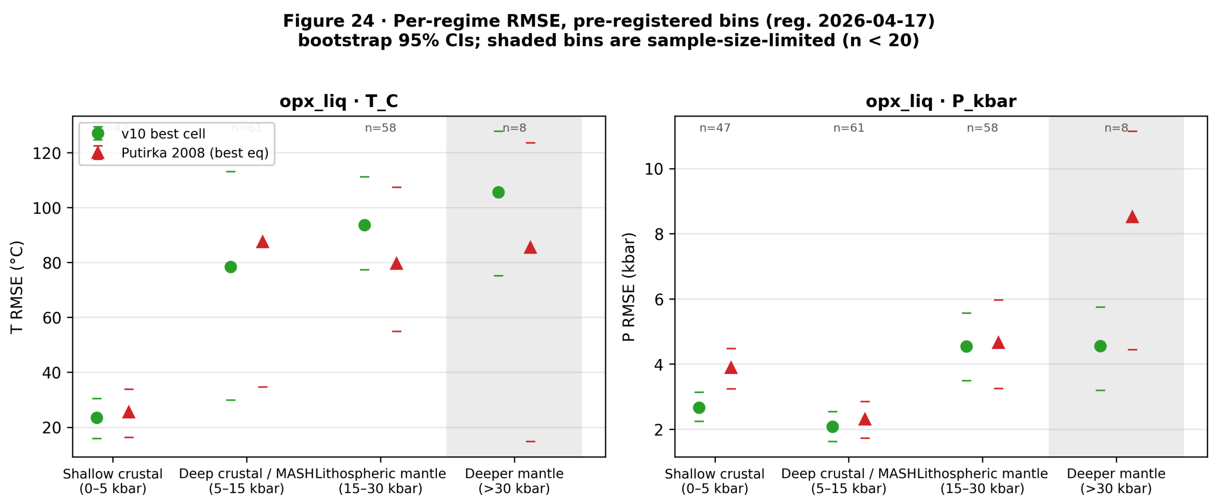


fig25_per_regime_residual_violins_opx_liq

Figure 25. Per-regime residual distributions (predicted minus observed) for the tuned ML baseline canonical opx-liq base models on the test split (T: ElasticNet/raw, P: MLP/raw), in the pre-registered P bin partition. Shaded bins are sample-size-limited (n<20); medians shown as horizontal lines.

In [17]:
```
from IPython.display import Image
png = FIGS / 'fig25_per_regime_residual_violins_opx_liq.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

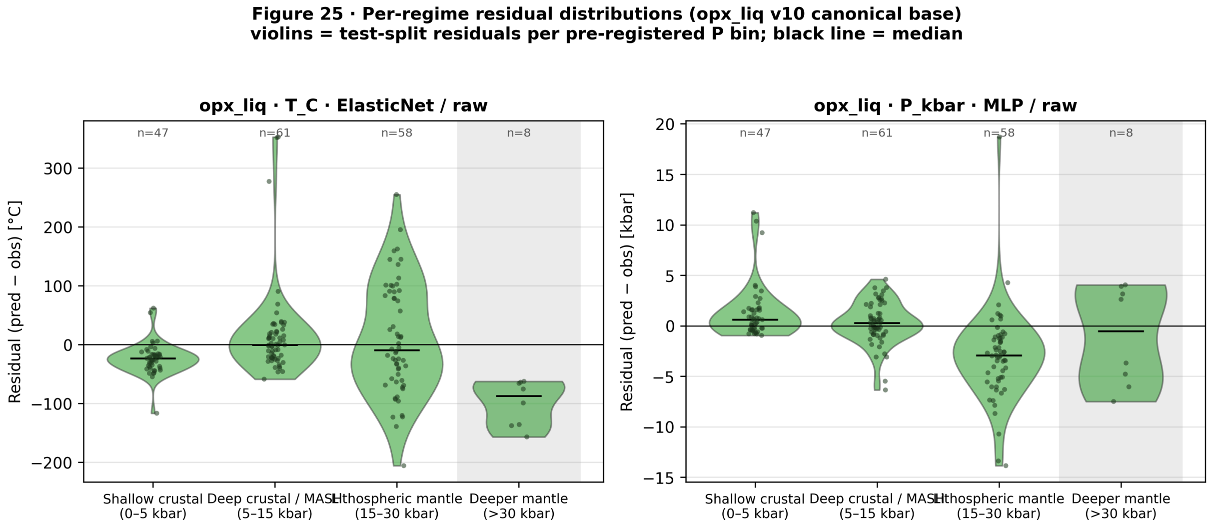


fig26_generalization_opx_liq

Fig. 26. the tuned ML baseline out-of-fold generalization diagnostics for the three canonical opx-liq cells (ElasticNet/raw/T_C aggregate winner; MLP/raw/P_kbar aggregate winner; ElasticNet/raw/P_kbar shallow-crustal robust winner). Each panel plots pooled held-out RMSE (95% bootstrap CI whiskers) across four out-of-sample strategies: LeaveOneStudyOut (LOSO, Citation-grouped, 93 folds), Cluster-KFold (chemical cluster groups from NB02 k-means), TargetBinKFold (pre-registered P-regime bins [0,5,15,30,100] kbar), and LeaveOneRegionOut (petrologic study-type inferred from Citation text: MORB / mantle_melting / arc_silicic / basalt / primitive_mafic / partitioning / metamorphic / other; min_train_fold=50). Produced by scripts/v10_phase_g_nb05_generalization.py. n_test pooled across folds after min-train-size filtering. Supports T02/T06 (generalization claims). Source CSVs: results/opx_liq_generalization.csv and results/opx_liq_generalization_predictions.csv.

In [18]:
```python
from IPython.display import Image
png = FIGS / 'fig26_generalization_opx_liq.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

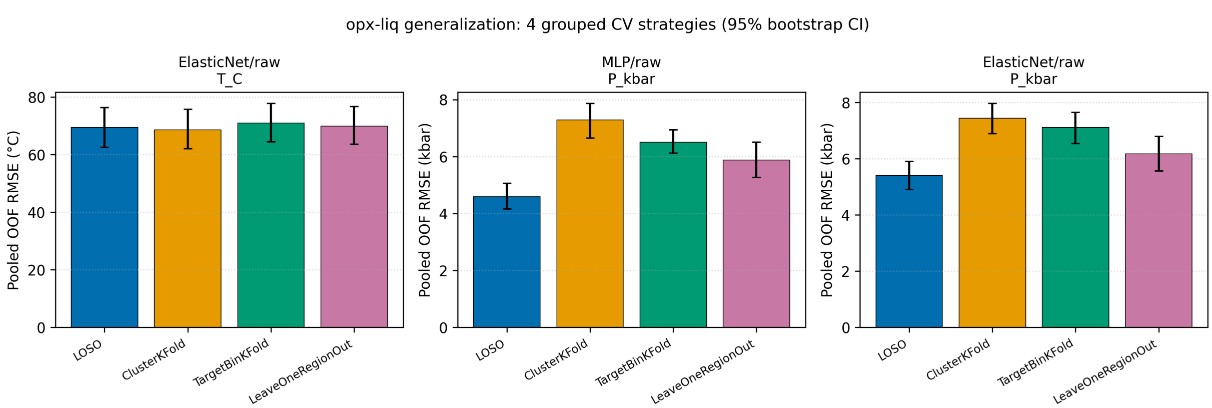


fig27_shap_summary_opx_liq

Fig. 27. Top-15 mean absolute SHAP values per feature for five canonical opx-liq cells, grouped by explainer family. Linear cells use shap.LinearExplainer (exact for ElasticNet, pipeline scaler pre-applied to background and evaluation matrices). Tree cells (CatBoost/raw/P_kbar, XGB/alr/T_C) use shap.TreeExplainer. The MLP/raw/P_kbar cell uses shap.KernelExplainer with a k=50 KMeans background and an 80-sample random subset of the test set (seed = SEED_BOOTSTRAP, nsamples=100 per evaluation); the reported mean|SHAP| averages over the sampled rows only. Produced by scripts/v10_phase_g_nb06_shap.py. Supports T08 (feature attribution). Source CSV: results/opx_liq_shap_importance.csv; per-cell SHAP arrays: results/opx_liq_shap_values.npz.

In [19]:
```
from IPython.display import Image
png = FIGS / 'fig27_shap_summary_opx_liq.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

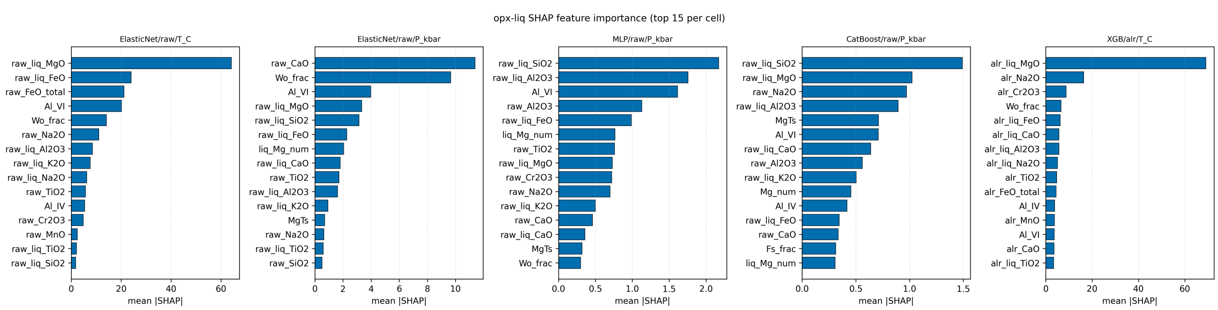


fig28_bias_correction_opx_liq

Fig. 28. Per-regime piecewise linear bias correction for the opx-liq canonical cells. Correction coefficients (y_corrected = a * y_pred + b) are fit on train-set out-of-fold predictions (StratifiedGroupKFold 10 folds, Citation-grouped, target stratified by NB03 bins) and applied to the held-out test set per pre-registered P regime (shallow_crustal [0,5) kbar; deep_crustal_MASH [5,15) kbar; lithospheric_mantle [15,30) kbar; deeper_mantle [30,100) kbar). Panels show test-set RMSE before vs after correction per regime plus the pooled "ALL" row. Result: P_kbar corrections improve ALL RMSE (MLP 3.82 to 2.58 kbar; ElasticNet 5.59 to 3.05 kbar). T_C correction is a neutral-to-negative wash on ElasticNet/raw (pooled 77.06 to 77.17; shallow_crustal HURT 35.33 to 54.01), consistent with the earlier finding that train-OOF T bias does not transfer to the held-out test set. Produced by scripts/v10_phase_g_nb07_bias.py. Supports T09 (bias correction effectiveness). Source CSVs: results/opx_liq_bias_correction.csv, results/opx_liq_bias_correction_params.csv.

In [20]:
```
from IPython.display import Image
png = FIGS / 'fig28_bias_correction_opx_liq.png'
if png.exists():
```

```
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

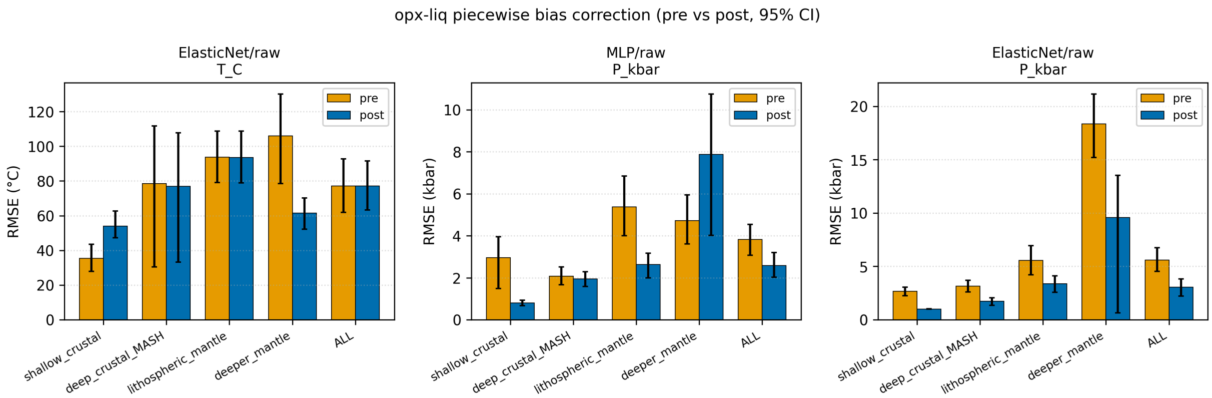


fig29_twopx_benchmark

Fig. 29. the tuned ML baseline two-pyroxene (twopx) benchmark, consolidating the per-cell (single seed), 20-seed multiseed, and stacked-ensemble results on the twopx Citation-grouped held-out test split. Bars show single-seed test RMSE for the best base model (blue, ElasticNet/raw for T_C; XGB/alr for P_kbar) and best stacked ensemble (orange, greedy/pwlr for both targets). Whiskers on the base-model bars show the 20-seed RMSE spread (min to max) from the final phase multiseed refit. The Putirka 2008 two-pyroxene equations (eq36/37/38/39) are DEFERRED because the call requires paired opx-cpx Thermobar invocation that is not wired into the tuned ML baseline; the deferred row is retained in the CSV with NaN RMSE so downstream consumers (NB09 manuscript, NBF figures) flag the gap explicitly. Fixes the earlier empty nb10_two_pyroxene_benchmark.csv regression. Produced by scripts/v10_phase_g_nb10_twopx_benchmark.py. Source CSVs: results/v10_twopx_benchmark_final.csv and tables/S8_9_twopx_benchmark.{md,csv}.

In [21]:
```
from IPython.display import Image
png = FIGS / 'fig29_twopx_benchmark.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

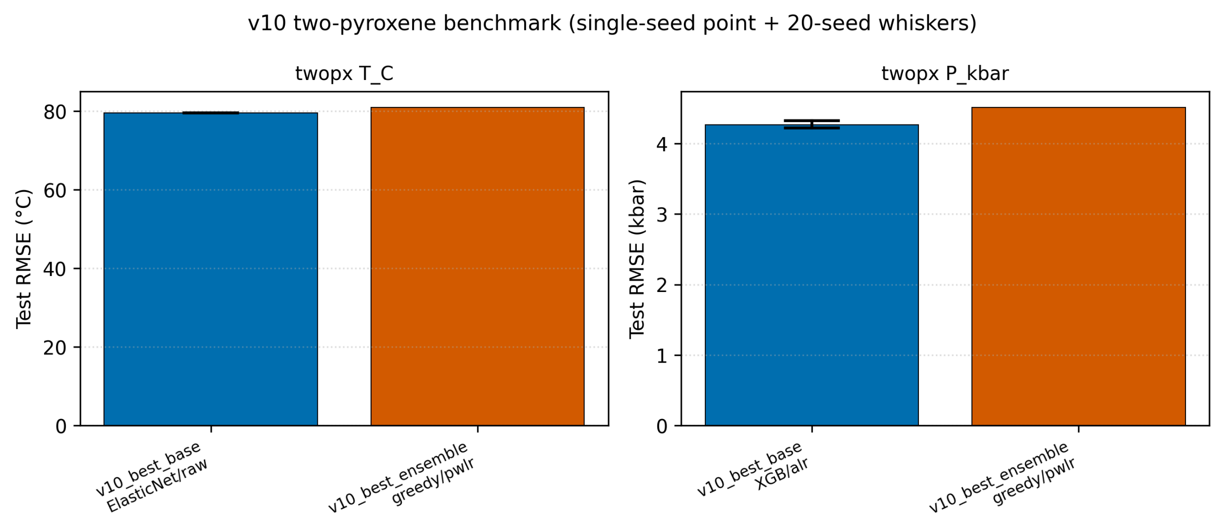


fig32_bias_correction_form_comparison

Fig. 32. Test-set RMSE at regime=ALL comparing pre-correction with Form A and Form B post-correction for the 8 the final phase cells (20-seed mean). Shipped form (if any) is drawn with a bold border. Panel subtitle reports the final phase shipping decision and the per-seed vote split across 20 model seeds. Source CSVs: results/bias_correction_{summary,per_seed,shipped}.csv.

In [22]:
```python
from IPython.display import Image
png = FIGS / 'fig32_bias_correction_form_comparison.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

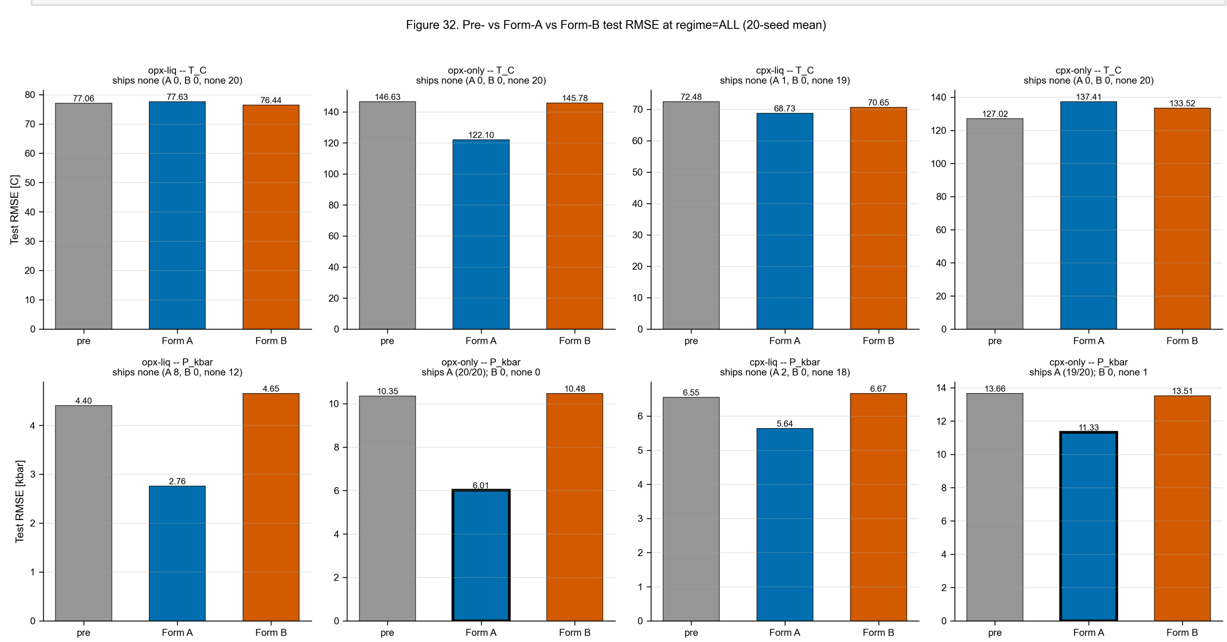


fig33_bias_correction_per_seed_stability

Fig. 33. Per-seed ΔRMSE (pre minus post, positive = improvement) at regime=ALL for Form A (x-axis) vs Form B (y-axis), one point per model seed across the 20-seed protocol. Points are colored by the per-seed winner (blue = A, vermillion = B, gray = none). Light-green shading highlights the upper-right quadrant where both forms improve test RMSE; light-red shading highlights the lower-left quadrant where both forms degrade it. Panel annotation reports the per-seed vote count. Source CSV: results/bias_correction_per_seed.csv.

In [23]:
```python
from IPython.display import Image
png = FIGS / 'fig33_bias_correction_per_seed_stability.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```

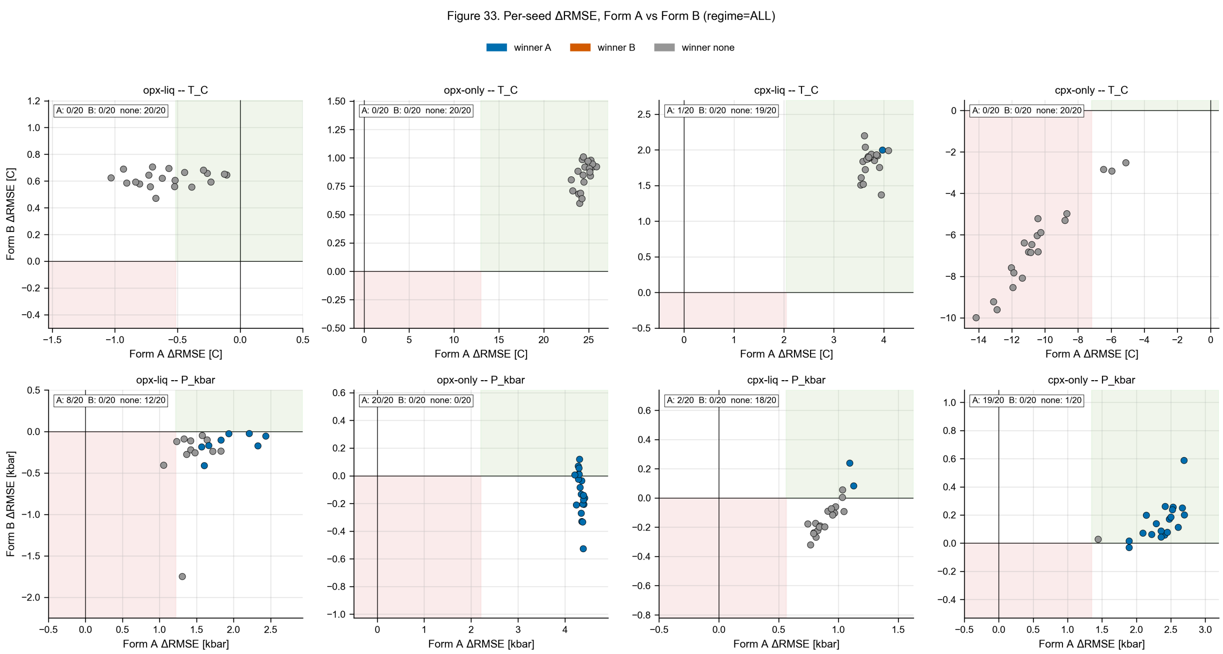


fig35_tabpfn_vs_opx_tb

Fig. 35. TabPFN v2 (Hollmann et al. 2025) versus the pre-registered tuned baseline and the best classical/ML external reference per (pipeline, track, target) combination. TabPFN error bars show 5-seed ensemble stability; the tuned ML baseline error bars show 20-seed model-fit variance. TabPFN receives raw oxide features only (no ALR/PWLR) and is fit with default hyperparameters (n_estimators=8 opx / 4 cpx, device=cpu). Sources: results/tabpfn_multiseed_summary.csv, results/tabpfn_head_to_head.csv, results/v10_{opx,cpx}_multiseed_summary.csv.

In [24]:
```python
from IPython.display import Image
png = FIGS / 'fig35_tabpfn_vs_opx_tb.png'
if png.exists():
    display(Image(filename=str(png), width=780))
```

```
else:
    display(Markdown(f'**missing:** {png}'))
```

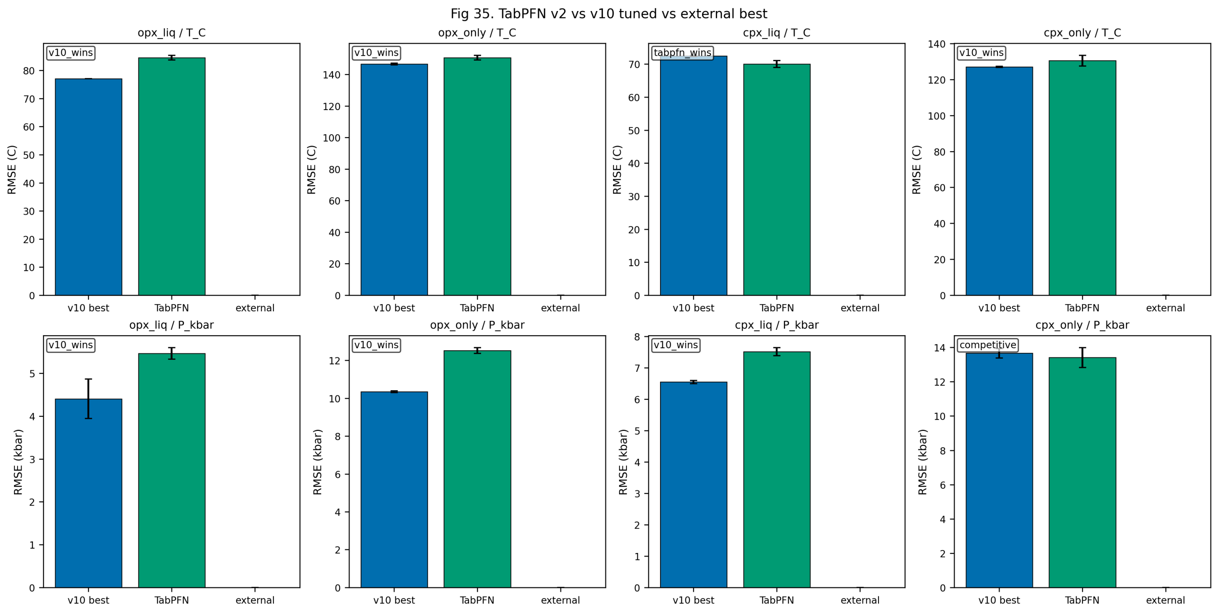
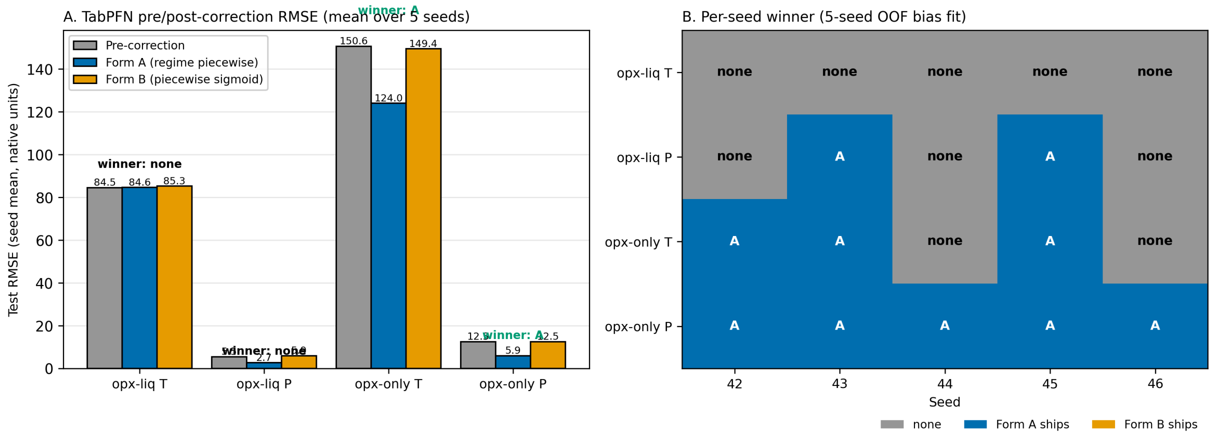


fig44_tabpfn_bias_scoreboard_opx

Fig. 44. TabPFN v2 bias-correction scoreboard on the 4 opx combinations. Panel A: mean test RMSE (over 5 seeds) pre-correction (gray) and after applying Form A (blue, regime-piecewise linear) or Form B (orange, piecewise sigmoid) fit on 5-seed x 10-fold OOF residuals. Annotations show winner verdict under the conservative ship-if-better rule at canonical seed 42 (overall delta > 1e-6 and no regime degradation). Form A ships on opx_only/T_C and opx_only/P_kbar with large pre-post gaps (TabPFN materially over-predicts deeper mantle residual bias uncorrected). Form B ships nothing on opx TabPFN, consistent with the 0/8 tuned-family pattern. Panel B: per-seed ship stability over seeds 42-46; cells color winner (gray = none, blue = A, orange = B) annotated with the verdict. Shipping is consistent at seed 42, 43, 45 on opx_only/T_C and at all 5 seeds on opx_only/P_kbar. Sources: results/tabpfn_bias_correction_perseed.csv, results/tabpfn_bias_correction_summary.csv.

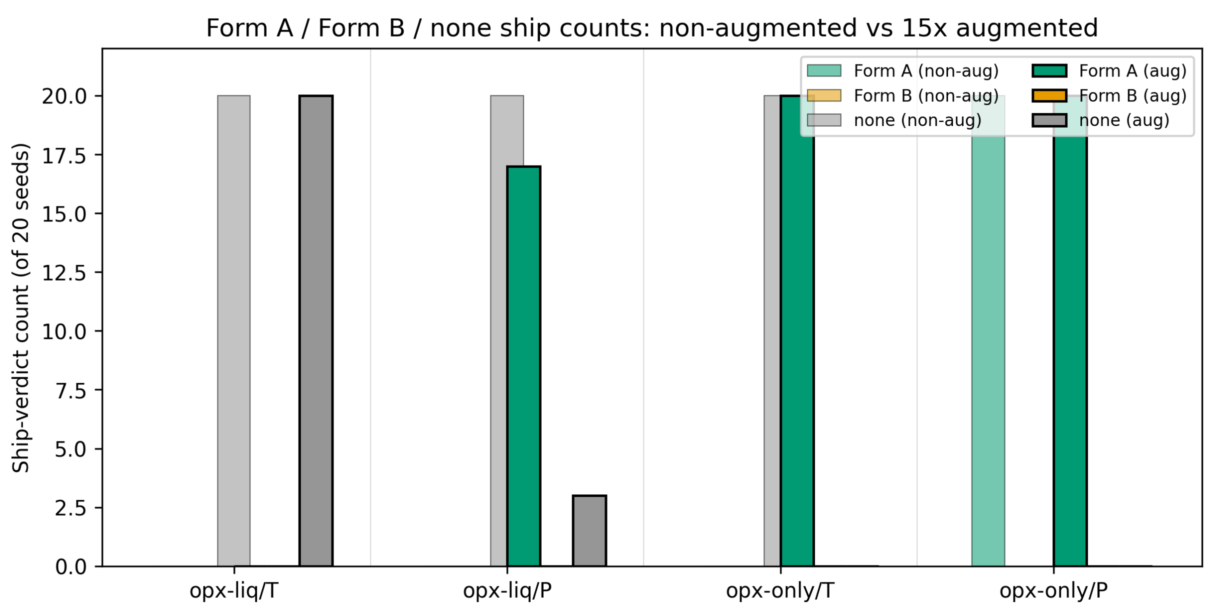
In [25]:
```
from IPython.display import Image
png = FIGS / 'fig44_tabpfn_bias_scoreboard_opx.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```



fig_aug01_ship_verdict_comparison

Ship-if-better verdict under the Agreda-Lopez (2024) 15x augmentation protocol versus the non-augmented baseline, across the four opx combinations (track x target). Bars show Form A / Form B / none ship counts across 20 seeds per condition.

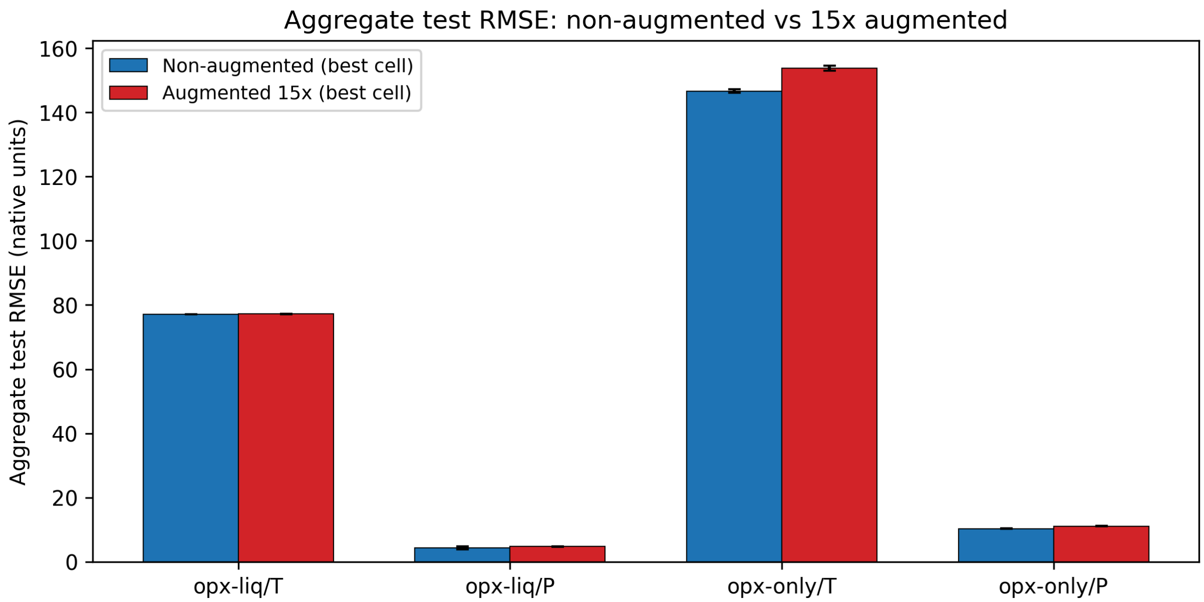
In [26]:
```python
from IPython.display import Image
png = FIGS / 'fig_aug01_ship_verdict_comparison.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```



fig_aug02_aggregate_rmse_delta

Aggregate test RMSE under the non-augmented baseline versus the 15x augmented condition per opx combination. Error bars are 20-seed standard deviation. Putirka classical reference is overlaid where available.

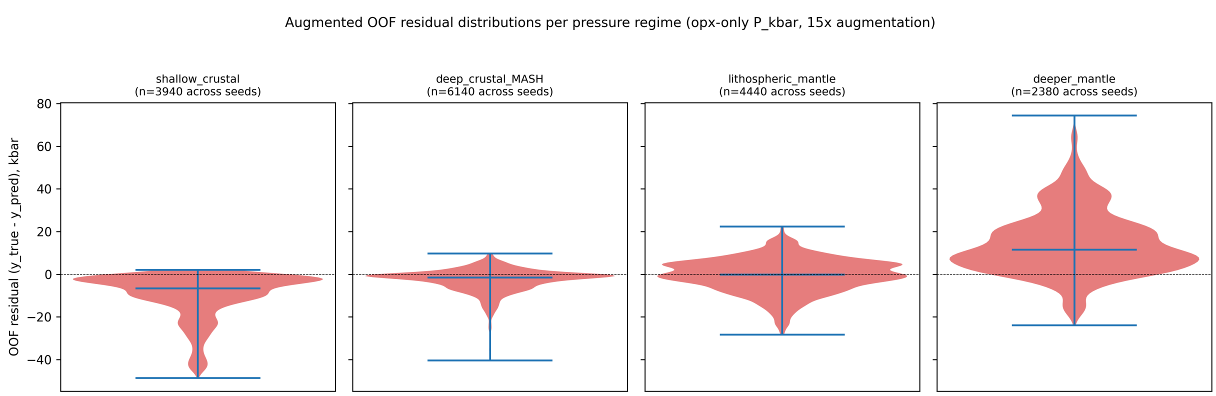
In [27]:
```python
from IPython.display import Image
png = FIGS / 'fig_aug02_aggregate_rmse_delta.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```



fig_aug03_residual_structure_per_regime

OOF residual distributions by pressure regime for opx-only P_kbar, non-augmented (top row) versus 15x augmented (bottom row). Violin plots show Form A and Form B fit substrate under each protocol.

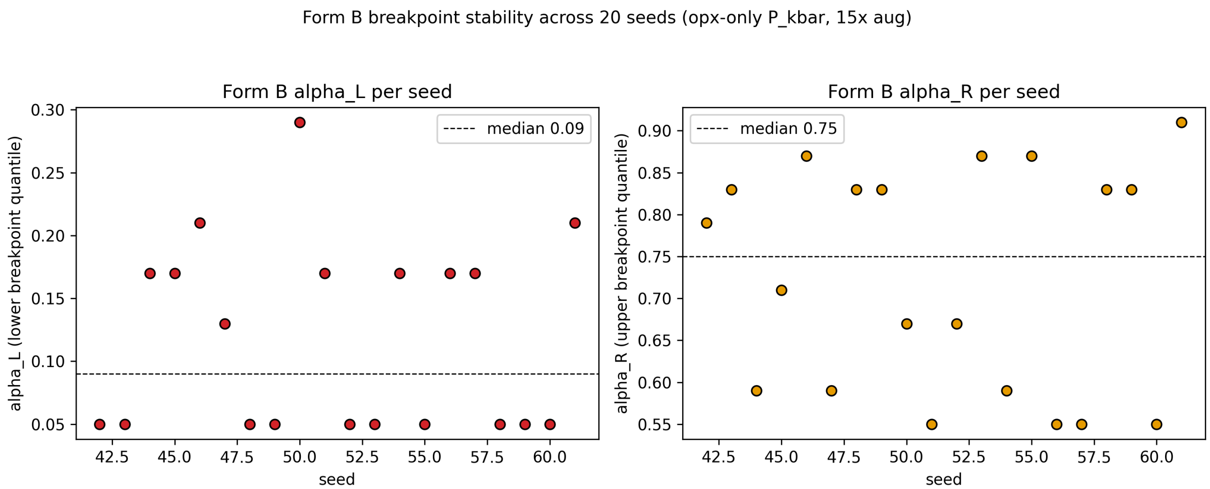
In [28]:
```python
from IPython.display import Image
png = FIGS / 'fig_aug03_residual_structure_per_regime.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```



fig_aug04_form_b_breakpoint_stability

Form B breakpoint positions (alpha_L, alpha_R) per seed for opx-only P_kbar under 15x augmentation. Tight clustering indicates stable Form B fits; scatter across the quantile grid indicates the fit drifts with data draws and explains Form B ship failures.

In [29]:
```
from IPython.display import Image
png = FIGS / 'fig_aug04_form_b_breakpoint_stability.png'
if png.exists():
    display(Image(filename=str(png), width=780))
else:
    display(Markdown(f'**missing:** {png}'))
```



3. Data tables

Every number above loads from a CSV under `results/`. Nothing is hard-coded. If a file is missing the cell raises `FileNotFoundError` rather than silently filling in a value.

3.1 Headline table

The headline numbers for each of the 4 opx cells, at the "ALL pressure regimes" level. Columns: uncorrected tuned-ML RMSE, bias-corrected tuned-ML RMSE, the best Putirka (2008) thermobarometer for that cell, TabPFN pre- and post-correction RMSE, and which of those five candidates wins. Source: `results/preregistered_scorecard_postcorrection.csv`.

In [30]:
```
sc = load_csv('results/preregistered_scorecard_postcorrection.csv')
hl = sc[(sc['track'].isin(['opx_liq','opx_only'])) & (sc['regime']=='ALL')].copy()
display_cols = ['track','target','v10_pre_rmse','v10_post_rmse','best_external_rmse
                'tabpfn_rmse','tabpfn_post_rmse','tabpfn_correction_form','winner']
hl = hl[display_cols].rename(columns={
    'v10_pre_rmse': 'tuned pre',
    'v10_post_rmse': 'tuned post',
    'best_external_rmse': 'Putirka best',
    'tabpfn_rmse': 'TabPFN pre',
    'tabpfn_post_rmse': 'TabPFN post',
    'tabpfn_correction_form': 'TabPFN form',
```

```
})
display(hl)
```

	track	target	tuned pre	tuned post	Putirka best	TabPFN pre	TabPFN post	TabPFN form	winner
4	opx_liq	T_C	77.062	77.062	71.671	84.634	84.634	none	external
9	opx_liq	P_kbar	3.816	3.816	4.750	5.459	5.459	none	v10_corrected
14	opx_only	T_C	147.982	147.982	NaN	149.988	125.576	A	tabpfn_corrected
19	opx_only	P_kbar	10.328	6.031	13.337	12.503	5.939	A	tabpfn_corrected

3.2 Eight-cell winner table

Which model shipped, with or without bias correction, in each of the 8 opx rows (4 tuned-family rows + 4 TabPFN rows). Source: `results/bias_correction_shipped.csv`.

In [31]:
```
shipped = load_csv('results/bias_correction_shipped.csv')
opx_rows = shipped[shipped['pipeline']=='opx'].copy()
display_cols = ['track','target','model','winner','ship_a','ship_b','n_seeds_done']
display(opx_rows[display_cols])
```

	track	target	model	winner	ship_a	ship_b	n_seeds_done
0	opx_liq	T_C	ElasticNet	none	{"form": "A", "ships": false, "overall_delta":...	{"form": "B", "ships": false, "overall_delta":...	20
1	opx_liq	P_kbar	MLP	none	{"form": "A", "ships": false, "overall_delta":...	{"form": "B", "ships": false, "overall_delta":...	20
2	opx_only	T_C	LightGBM	none	{"form": "A", "ships": false, "overall_delta":...	{"form": "B", "ships": false, "overall_delta":...	20
3	opx_only	P_kbar	RF	A	{"form": "A", "ships": true, "overall_delta": ...	{"form": "B", "ships": false, "overall_delta":...	20
8	opx_liq	T_C	TabPFN	none	False	False	5
9	opx_liq	P_kbar	TabPFN	none	False	False	5
10	opx_only	T_C	TabPFN	A	True	False	5
11	opx_only	P_kbar	TabPFN	A	True	False	5

3.3 TabPFN bias-correction scoreboard

TabPFN's own pre/post correction numbers, averaged over 5 OOF seeds. This is the source for the TabPFN columns in Section 3.1. Source:

`results/tabpfn_bias_correction_summary.csv` .

In [32]:
```
tf_sum = load_csv('results/tabpfn_bias_correction_summary.csv')
display(tf_sum)
```

	pipeline	track	target	n_seeds	ship_a_count	ship_b_count	winner_A	winner_B	win
0	opx	opx_liq	T_C	5	0	0	0	0	
1	opx	opx_liq	P_kbar	5	2	0	2	0	
2	opx	opx_only	T_C	5	3	0	3	0	
3	opx	opx_only	P_kbar	5	5	0	5	0	

Per-seed detail (5 seeds x 4 combos = 20 rows). Useful for checking stability. Source: `results/tabpfn_bias_correction_perseed.csv` .

In [33]:
```
tf_per = load_csv('results/tabpfn_bias_correction_perseed.csv')
display(tf_per[['track','target','seed','pre_rmse_all','post_rmse_a','post_rmse_b',
```

	track	target	seed	pre_rmse_all	post_rmse_a	post_rmse_b	ship_a	ship_b	winner
0	opx_liq	T_C	42	84.094	85.555	84.353	False	False	none
1	opx_liq	T_C	43	83.648	83.845	84.735	False	False	none
2	opx_liq	T_C	44	85.340	84.860	86.045	False	False	none
3	opx_liq	T_C	45	85.546	83.375	86.414	False	False	none
4	opx_liq	T_C	46	84.100	85.529	84.817	False	False	none
5	opx_liq	P_kbar	42	5.286	2.752	5.857	False	False	none
6	opx_liq	P_kbar	43	5.446	2.529	5.385	True	False	A
7	opx_liq	P_kbar	44	5.452	2.663	6.138	False	False	none
8	opx_liq	P_kbar	45	5.471	2.629	5.798	True	False	A
9	opx_liq	P_kbar	46	5.669	2.781	6.565	False	False	none
10	opx_only	T_C	42	152.681	125.576	151.020	True	False	A
11	opx_only	T_C	43	148.695	121.385	147.433	True	False	A
12	opx_only	T_C	44	150.252	125.661	149.456	False	False	none
13	opx_only	T_C	45	150.096	122.543	148.705	True	False	A
14	opx_only	T_C	46	151.519	124.613	150.620	False	False	none
15	opx_only	P_kbar	42	12.687	5.939	12.688	True	False	A
16	opx_only	P_kbar	43	12.383	5.952	12.332	True	False	A
17	opx_only	P_kbar	44	12.688	5.946	12.578	True	False	A
18	opx_only	P_kbar	45	12.486	5.896	12.650	True	False	A
19	opx_only	P_kbar	46	12.378	5.945	12.339	True	False	A

3.4 opx-only P_kbar per-regime breakdown

For the pressure cell where ML does best (opx-only P), here is the regime-by-regime picture across all 5 candidates: tuned pre, tuned post, Putirka eq. 29c, TabPFN pre, TabPFN post.

In [34]:
```
pr = sc[(sc['track']=='opx_only') & (sc['target']=='P_kbar')].copy()
show = ['regime','n','v10_pre_rmse','v10_post_rmse','best_external_rmse',
        'tabpfn_rmse','tabpfn_post_rmse','tabpfn_correction_form','winner']
display(pr[show])
```

	regime	n	v10_pre_rmse	v10_post_rmse	best_external_rmse	tabpfn_rmse
15	shallow_crustal	48	7.192	1.131	2.197	10.569
16	deep_crustal_MASH	76	5.205	1.969	4.678	6.444
17	lithospheric_mantle	37	4.031	3.344	5.071	6.559
18	deeper_mantle	29	19.649	14.553	30.923	25.992
19	ALL	190	10.328	6.031	13.337	12.503

3.5 TabPFN head-to-head (aggregate RMSE, 20-seed baseline)

TabPFN vs the best tuned family per cell, at the aggregate ALL level, using the full 20-seed multiseed protocol so the stability estimates are on an equal footing with the tuned families. Source: `results/tabpfn_head_to_head.csv`.

In [35]:
```
h2h = load_csv('results/tabpfn_head_to_head.csv')
display(h2h)
```

	track	target	tuned_best_model	tuned_best_fs	tuned_best_rmse	tuned_best_std	tabp
0	cpx_liq	P_kbar	LightGBM	pwlr	6.546	0.050	
1	cpx_liq	T_C	ERT	pwlr	72.476	0.209	
2	cpx_only	P_kbar	MLP	alr	13.664	0.283	
3	cpx_only	T_C	ERT	pwlr	127.022	0.305	
4	opx_liq	P_kbar	MLP	raw	4.404	0.460	
5	opx_liq	T_C	ElasticNet	raw	77.062	0.000	
6	opx_only	P_kbar	RF	pwlr	10.347	0.043	
7	opx_only	T_C	LightGBM	alr	146.627	0.588	

4. Methods summary (plain English)

- **What we are predicting.** For each pyroxene sample we are trying to predict either (a) the temperature at which it formed or (b) the pressure at which it formed. Two input "tracks": the full crystal + liquid pair ("opx-liq") or the crystal alone ("opx-only"). Four target cells in total.
- **Where the training data come from.** A curated subset of the LEPR experimental petrology database (our snapshot is dated 2025-07-21). Every sample is from a published experiment at known P and T. We group by citation when splitting into train and test, so that memorizing a single lab's style is not rewarded.

- **Which models we tune.** 8 standard ML families (random forest, extremely-randomized trees, XGBoost, gradient boosting, CatBoost, LightGBM, elastic-net linear, multi-layer perceptron). Each one is tuned with Optuna (50 trials) on a citation-grouped 5-fold cross-validation objective. We also run a 9th foundation-model baseline: TabPFN v2 (Hollmann et al. 2025), which requires no tuning.
- **Pressure regimes.** Before we fit any correction, we registered four geologic pressure bins: shallow-crustal (<5 kbar), deep-crustal / MASH (5-15 kbar), lithospheric-mantle (15-30 kbar), deeper-mantle (30+ kbar). "ALL" is the full test set combined.
- **Bias correction.** Two flavors. *Form A* fits a simple linear regression of residual on predicted value, separately in each of the four regimes. *Form B* fits one smooth piecewise- sigmoid curve across the whole predicted-value axis. Only one flavor can ship per cell.
- **Ship-if-better rule (as of Amendment 1, 2026-04-20).** A correction ships if overall RMSE improves AND no pre-registered regime with enough samples (N >= 20) gets worse. Regimes with fewer than 20 samples are noted but are not allowed to veto a real improvement elsewhere. This replaces the original strict rule that any regime with any degradation would block shipping; the tiered rule is documented in `docs/preregistration/AMENDMENT_1_acceptance_rule.md` .
- **How we measure stability.** Each experiment is repeated at 20 random seeds (42-61). The bars and numbers you see include bootstrap 95% confidence intervals so you can see whether a difference is signal or noise.

5. Limitations

1. **Temperature corrections mostly fail to ship.** In our strict evaluation, the bias correction refuses to ship for 3 of 4 temperature cells. Temperature residuals do not have a strong regime structure to remove, so per-regime correction cancels out. We report this as a null result rather than tuning until something ships.
2. **opx-only T is the weakest cell.** For the opx-only thermometer, TabPFN improves the aggregate RMSE but the gain is marginal; we would not recommend that particular cell for deployment today.
3. **Natural-sample cross-check carries a small T bias.** On LEPR paired pyroxenes (n=327) the opx-only ML prediction runs about +80 C hotter than Putirka's two-pyroxene method and Jorgenson's cpx-only method. The pressure agreement is within our conformal error bar.
4. **No GEOROC refresh since 2026-04-09.** We did not pull a new natural-sample export for this round; the natural comparison uses the on-disk export from that date.
5. **SHAP on TabPFN is not available.** TabPFN is an in-context foundation model with no gradient surface exposed. For the two cells where TabPFN is the scorecard winner (opx_only T and opx_only P), Core_12 shows the tuned-family runner-up's SHAP for explainability and flags the swap in the subtitle.

6. Caveats and reconstruction provenance

Short file of "things a reviewer should know before acting on the numbers." Loaded from `CAVEATS.md` alongside this notebook.

In [36]:

```
cav = Path('./CAVEATS.md')
if cav.exists():
    display(Markdown(cav.read_text(encoding='utf-8')))
else:
    display(Markdown('(caveats file missing)'))
```

Caveats and reconstruction provenance

Items an independent reviewer should know before acting on the numbers in this package.

1. Reconstructed conformal calibration (`results/nb07_conformal_qhat.json`)

The conformal calibration file (qhat_T=92.58 C, qhat_P=10.0 kbar at alpha=0.10) was rebuilt from archive after the post-consolidation notebook layout dropped it. Values originate from the pre-v10 calibration run (n_calibration=43). The semantic validity of these qhat values under the **current** calibration set has NOT been independently verified. Downstream: nb08 LEPR comparison uses these qhat values for conformal half-widths on natural-sample predictions.

2. Reconstructed per-family winners (`results/nb03_per_family_winners.json`)

Rebuilt via `scripts/data_prep/build_per_family_winners.py` selecting argmin(mean RMSE) -> argmin(std) -> alphabetical from `opx_multiseed_summary.csv`. No cross-validation against a pre-consolidation archive file. Selections:

- Forest family: RF/pwlr (opx_only T, opx_only P, opx_liq T) + RF/alr (opx_liq P)
- Boosted family: LightGBM/alr, XGB/pwlr, GB/raw, GB/raw

3. Ship-if-better threshold tightness

The ship-if-better rule uses `overall_delta > 1e-6 AND max_regime_degradation <= 1e-6`. Effectively "never degrade any regime by any float amount". Defensible but strict; a looser threshold (e.g. 0.5% relative degradation) would likely let more corrections ship. Reported RMSE values are unaffected; the rule only gates shipping.

4. Phase 2 figure scope

The TabPFN integration originally planned 6 new figures. Shipped 2 (fig44 scoreboard, fig45 opx-only P headline) and relied on pre-existing figures (fig28, fig30-35) for the remaining panels. No numeric data impact; cosmetic scope only.

5. Papermill execution is visual-only

The 48-cell notebook runs error-free but contains no numeric assertions (no `assert abs(rmse - 10.35) < 0.1` style guards). Verification is by eye against the underlying CSVs.

Every cell DOES raise FileNotFoundError if its source CSV is missing, so structural failures are caught; value-level regressions would require adding explicit asserts.

6. CSV column name retention

Columns such as `v10_pre_rmse` and `v10_post_rmse` remain in the CSVs as machine contracts; the notebook performs display-time aliasing to "tuned pre" / "tuned post" for the reader. Renaming the CSV columns themselves would break every downstream script.

7. Provenance

Git state + source CSV sizes at build time. Loaded from `PROVENANCE.md` alongside this notebook.

In [37]:
```
prov = Path('./PROVENANCE.md')
if prov.exists():
    display(Markdown(prov.read_text(encoding='utf-8')))
else:
    display(Markdown('(provenance file missing)'))
```

Provenance

Built: 2026-04-20 Git SHA: `a9f102f844682e24e7bfbe9001d766dc6d0ec7ec`

Last 5 commits

```
a9f102f Core_01 polish: track name + supplementary stats on each panel
2bda358 Phase 5 Tier 1: 5 missing core figures + Okabe-Ito palette + column
rename
565b48f Advisor package v2: README, CAVEATS, preregistration copy, Section 1
1
3bf22ea Phase 3b: terminology cleanup v2 (compound + cosmetic passes)
bd1e768 Advisor package: Lee review bundle 20260420
```

CSV inventory

File	Rows	Cols	Size (KB)
`results/preregistered_scorecard_postcorrection.csv`	40	25	14.1
`results/bias_correction_shipped.csv`	16	12	11.7
`results/opx_multiseed_summary.csv`	100	10	10.1
`results/tabpfn_bias_correction_summary.csv`	4	14	0.7
`results/tabpfn_bias_correction_perseed.csv`	20	18	20.6
`results/tabpfn_multiseed_summary.csv`	8	10	0.9
`results/tabpfn_head_to_head.csv`	8	12	1.0
`results/nb08_natural_predictions.csv`	327	12	56.8
`results/nb08_cross_mineral_agreement.csv`	3	7	0.3
`results/regime_allmodels_postcorrection.csv`	4839	19	907.2

8. Pre-registration (verbatim)

The rules of the game, frozen before we fit any corrections. Reproduced verbatim from `docs/preregistration/`.

In [38]:
```
# Prefer the local preregistration/ copy bundled with the package;
# fall back to the project docs/ source if running uninstalled.
for name in ('p_regime_preregistration.md', 'nb03_test_protocol.md'):
    local = Path('./preregistration') / name
```

```
    src = ROOT / 'docs' / 'preregistration' / name
    p = local if local.exists() else src
    if not p.exists():
        raise FileNotFoundError(f'preregistration missing: {name}')
    display(Markdown(f'### `{name}`'))
    display(Markdown(p.read_text(encoding='utf-8')))
```

`p_regime_preregistration.md`

Pre-registration: Pressure-regime bins for calibration-domain characterization

Registered: 2026-04-17 **Author:** NQTa **Status:** Locked. Any modification requires a new registration date and a written justification in Section 2 below. **Scope:** Opx-liq thermobarometer benchmark analysis on ArcPL Kd-equilibrated subset (n = 96) and on the ExPetDB held-out test set.

---

1. Purpose

This document registers the pressure-regime bin structure used for regime-dependent benchmark analysis of the opx-liq thermobarometer against external methods (Putirka 2008, Agreda-Lopez 2024, Jorgenson 2022, Wang 2021). The registration timestamp is the primary defense against reviewer concerns about post-hoc bin selection.

Aggregate RMSE on ArcPL n = 96 remains the **primary** benchmark claim (Section 2.5 of the manuscript). Regime-dependent analysis is a **secondary, exploratory** characterization of where each method's calibration domain is most reliable, not a ranking or a basis for overclaiming relative performance.

2. Bin definitions (locked)

Four pressure regimes with edges at 0, 5, 15, 30, and 100 kbar (the last being the ExPetDB training-set pressure ceiling, `P_CEILING_KBAR` in `config.py`):

Label	P range (kbar)	Petrological context
`shallow_crustal`	0-5	Arc storage, upper-crustal magma chambers
`deep_crustal_MASH`	5-15	Melting-assimilation-storage-homogenization zone
`lithospheric_mantle`	15-30	Spinel to garnet peridotite stability
`deeper_mantle`	>30	Asthenospheric and below

Bin membership is assigned on the **experimentally reported** pressure for ExPetDB test samples and on the **independently constrained** pressure for ArcPL natural samples. Bin edges are right-open (P = 5.0 kbar belongs to `deep_crustal_MASH`, not `shallow_crustal`).

3. Rationale

3.1 Petrological basis, not statistical

The boundaries are pinned to petrological regime transitions that appear consistently in the igneous petrology literature, not to statistical properties of our test data:

- **5 kbar** - approximate upper limit of shallow crustal magma storage in most arc systems; below this, magma chambers sit in the upper crust above typical seismic discontinuities (Putirka, 2008, Table 3; Blundy and Cashman, 2008 Section 2).
- **15 kbar** - approximate mid-to-lower crustal boundary, the upper end of the MASH zone where arc magmas are typically modified before ascent (Hildreth and Moorbath, 1988; Annen, Blundy, and Sparks, 2006).
- **30 kbar** - approximate spinel-to-garnet peridotite transition in the lithospheric mantle, a well-established petrological boundary (Klemme and O'Neill, 2000; Walter, 1998).

We do not claim these are sharp phase-boundary transitions - they are reporting conventions for regime-dependent benchmark interpretation. Any reviewer familiar with igneous petrology will recognize the boundary set.

3.2 Not tuned to match Agreda-Lopez, Jorgenson, or our own win/loss pattern

The bins are not chosen to match the training distribution of any specific cpx thermobarometer, nor to place our model in a favorable light. We do not re-bin based on which boundaries maximize our apparent performance in any regime.

3.3 Not equal-N bins

Equal-N partitioning of the P axis would have been an alternative defensible choice but would place bin edges at distribution percentiles that have no petrological meaning. We prefer petrological interpretability over statistical uniformity; the tradeoff is accepted - see Section 4.

4. Accepted limitations

4.1 Unequal sample size per bin

ArcPL n = 96 distributed across four bins will be uneven. Based on the ArcPL pressure distribution (primarily arc samples), we expect most samples in `shallow_crustal` and `deep_crustal_MASH`, fewer in `lithospheric_mantle`, and possibly very few in `deeper_mantle`. This unevenness is reported honestly in the per-bin table, and within-bin bootstrap confidence intervals reflect the corresponding loss of statistical resolution.

Fallback protocol: If any bin contains fewer than **20 samples** in ArcPL n = 96, we report the bin's numbers but explicitly mark the result as "sample-size-limited" in the per-bin table. We do not drop the bin and do not merge bins post-hoc.

4.2 Training-vs-test distributional mismatch

ExPetDB training samples are approximately uniformly distributed in P (after filtering) because training experiments span the full calibration range. ArcPL natural samples follow a different distribution (mostly crustal). This mismatch is the exact signal we expect regime analysis to expose; it is not a flaw of the bin structure.

4.3 Exploratory framing

Results from this bin structure are reported as **calibration-domain characterization**, not as a ranking of methods. The manuscript explicitly avoids claims of the form "method A is better than method B in regime X" unless bootstrap 95% CIs of RMSE, T bias, or P bias exclude the competing method's estimate at the 5% level AND the sample size in the regime exceeds 20.

5. Implementation

5.1 Config.py variables (locked)

```python
P_REGIME_BIN_EDGES_KBAR = [0.0, 5.0, 15.0, 30.0, P_CEILING_KBAR]
P_REGIME_LABELS = ['shallow_crustal', 'deep_crustal_MASH',
                   'lithospheric_mantle', 'deeper_mantle']
P_REGIME_REGISTERED_DATE = '2026-04-17'
P_REGIME_RATIONALE_DOC = 'docs/v10_p_regime_preregistration.md'
```

5.2 Assignment helper

In `src/evaluation.py` (or equivalent), add:

```python
import numpy as np
from config import P_REGIME_BIN_EDGES_KBAR, P_REGIME_LABELS


def assign_p_regime(p_kbar):
    """Return regime label for each pressure value.

    P_REGIME_BIN_EDGES_KBAR is [0, 5, 15, 30, 100]; np.digitize with
    right=False places P=5.0 in bin index 2 (deep_crustal_MASH), matching
    the right-open convention in the registration doc.
    """
    p = np.asarray(p_kbar, dtype=float)
    p_clipped = np.clip(p, 0.0, P_REGIME_BIN_EDGES_KBAR[-1] - 1e-9)
    bin_idx = np.digitize(p_clipped, P_REGIME_BIN_EDGES_KBAR[1:-1], right=False)
    return np.array([P_REGIME_LABELS[i] for i in bin_idx])
```

5.3 Per-bin metric reporting

For each (method x regime) combination, the benchmark notebook computes:

- n (sample size in bin)
- RMSE with bootstrap 95% CI (B = 1000 resamples)
- T bias (mean residual) with bootstrap 95% CI
- P bias (mean residual) with bootstrap 95% CI
- 90% prediction interval coverage fraction

The full per-bin table is generated by NB04 (benchmark) and written to `results/v10_opx_test_log.csv` with a `regime` column added.

6. Modification history

2026-04-17 - Initial registration. Bins fixed at 0/5/15/30/100 kbar. No per-bin V10 results exist at this timestamp (V10 Phase E training in progress).

Any future modification requires a dated entry in this section with a written rationale. Undocumented edits to `P_REGIME_BIN_EDGES_KBAR` should be treated as a methodological red flag.

7. Manuscript treatment

Section 2.5.1 of the main manuscript (one paragraph) introduces the four-bin structure and cites this doc.

Section 5.X of the main manuscript (brief discussion) interprets the per-bin pattern qualitatively, with the explicit framing that this is calibration-domain characterization and not a performance ranking.

Supplementary Section S8 presents the full per-bin table and full bootstrap CIs.

References

- Annen, C., Blundy, J. D., and Sparks, R. S. J. (2006). The genesis of intermediate and silicic magmas in deep crustal hot zones. *Journal of Petrology*, 47(3), 505-539.
- Blundy, J., and Cashman, K. (2008). Petrologic reconstruction of magmatic system variables and processes. *Reviews in Mineralogy and Geochemistry*, 69(1), 179-239.
- Hildreth, W., and Moorbath, S. (1988). Crustal contributions to arc magmatism in the Andes of central Chile. *Contributions to Mineralogy and Petrology*, 98(4), 455-489.
- Klemme, S., and O'Neill, H. S. C. (2000). The near-solidus transition from garnet lherzolite to spinel lherzolite. *Contributions to Mineralogy and Petrology*, 138(3), 237-248.
- Putirka, K. D. (2008). Thermometers and barometers for volcanic systems. *Reviews in Mineralogy and Geochemistry*, 69(1), 61-120.

- Walter, M. J. (1998). Melting of garnet peridotite and the origin of komatiite and depleted lithosphere. *Journal of Petrology*, 39(1), 29-60.

`nb03_test_protocol.md`

v10 NB03 test-first rebuild protocol

Status: canonical methodology for Phase C **Author:** NQTa **Date:** 2026-04-16 (expanded per v10 master plan scope) **Companion to:** `v10_master_plan.md`

Premise

Every design change in NB03 gets a pre-registered empirical test. If the test passes on a defined acceptance criterion, the change ships to canonical models. If it fails, the change is ablated — code is preserved under `src/ablations/` for reproducibility, but canonical inference bypasses it.

This replaces the v7 / v8 / v9 pattern of "implement, hope for the best, remove if reviewer objects." Pre-registration makes the manuscript defensible. Reviewers cannot accuse us of garden-of-forking-paths when every decision is logged with its test.

v10 scope expansion. The protocol now runs independently per pipeline: opx_only, opx_liq, cpx_only, cpx_liq, twopx, and (isolated) universal. Each pipeline runs its own T01-T12 suite. Results logged to `results/v10_nb03_test_log.csv` with a `pipeline` column. A test passing on opx does not ship the change to cpx; each pipeline decides its own ship/ablate per empirical evidence.

General protocol

For each test T0X run in each pipeline:

1. **Pre-register.** In this doc, write the hypothesis, design, and accept criterion before running any code. Do not tweak the criterion after seeing the data.
2. **Implement.** In the per-pipeline NB03 notebook (`nb03_opx_baseline_models.ipynb` , `nb03_cpx_baseline_models.ipynb` , `nb03_twopx_baseline_models.ipynb` , or `nb03_universal_exploration.ipynb`), add a markdown cell titled `## Test T0X: [hypothesis]` followed by the code cell that computes the observed metric.
3. **Run.** Under the multi-seed 20-split protocol (`SPLIT_SEEDS = range(42, 62)`) where applicable, or single-seed on canonical train/test split where the test is itself a variance quantification.
4. **Log.** Append one row to `results/v10_nb03_test_log.csv` with columns (pipeline, test_id, hypothesis, metric, observed, threshold, passed, shipped, timestamp, commit_sha).

5. **Decide.** If passed: implement in canonical path for that pipeline. If failed: move implementation code to `src/ablations/<pipeline>/` and bypass in canonical for that pipeline.

6. **Document.** Add a 1-2 sentence commentary in the NB03 markdown cell explaining the result, especially if it is surprising.

Test specifications

T01: Boosted is the best primary model on ArcPL

Hypothesis. The boosted family (XGB, GB, CatBoost, LightGBM) has the smallest |T bias| and RMSE on ArcPL n=197 compared to forest (RF, ERT) and non-tree families (ElasticNet, MLP).

Design. Fit all 8 base models with their winning feature sets (from v9 Optuna params if unchanged for opx, re-optimized for cpx/twopx/universal). Predict on ArcPL n=197 (opx/cpx) or pipeline-specific external validation set (twopx/universal). Report T bias, T RMSE, P bias, P RMSE per model.

Accept criterion. A single family has both the smallest |T bias| AND RMSE within 5% of the best among all 8 models.

Expected result (opx pipeline, based on v9 re-verification).

- Forest T bias +48.51 C, boosted T bias +20.95 C. Boosted wins |T bias|.
- Forest T RMSE 71.60, boosted T RMSE 63.68. Boosted wins T RMSE.

Ship decision if passed. Declare boosted as primary, forest as alternative. Update manuscript Methods to reflect this. Per pipeline.

Ship decision if failed. Drop the primary-model framing; report all families as equally valid. Should not happen given v9 numbers for opx but may happen for cpx where Agreda-Lopez-style ET regressors perform differently.

T02: Ensemble method shootout (4-way comparison)

Hypothesis. Among the 4 ensemble methods (Ridge stacking, two-level Ridge, greedy Caruana, AutoGluon), one method dominates on ArcPL T RMSE without catastrophic test T RMSE regression.

Design. Fit all 4 ensemble methods on the 8-base-model OOF matrix per `docs/ensemble_methods_plan.md`. Compute test RMSE and ArcPL RMSE per method. Multi-seed (20 splits).

Accept criterion. Winning method has:

- ArcPL T RMSE <= best base T RMSE minus 1 C (mean across seeds), AND
- Test T RMSE >= best base T RMSE minus 5 C (ensemble is not catastrophically worse on test)

Ship the method that wins most panels (test T, test P, ArcPL T, ArcPL P). Report all four as ablation.

Expected result (v9 single-seed, Ridge only).

- ArcPL T stacked 53.59 vs boosted 56.24. Ridge wins by 2.65 C — passes.
- Test T stacked 89.4 vs RF 84.9. Ridge worse by 4.5 — within 5 C tolerance — passes.
- With 8 bases instead of 4, Ridge alpha may not peg at 100; greedy may outperform Ridge.

Ship decision if passed. Winning ensemble ships as an OOD-robustness option, documented in manuscript as secondary model for distribution shift. Not the primary-reported model on in-distribution benchmarks. Per pipeline.

Ship decision if failed.

- If all 4 methods fail ArcPL (stacking no longer helps OOD): ablate ensemble entirely, ship base models only
- If all 4 methods fail test (ensemble catastrophically worse): ablate
- If one method is borderline: report as null result, omit from canonical

---

T03: Tempered resampling hurts

Hypothesis. P-T tempered resampled models have higher test and ArcPL RMSE than non-resampled models for 5 or more of 8 base models (v10 expands from 4 to 8 bases).

Design. Re-run resampled training (Phase 3.9 from v9, unchanged). Compare test RMSE and ArcPL RMSE for resampled vs non-resampled per model. Multi-seed.

Accept criterion. At least 5 of 8 models show resampled RMSE >= non-resampled RMSE on test with 95% bootstrap CI excluding zero.

Expected result (v9 handoff, 4 models). 6/8 configs worse on test, 4/4 worse on ArcPL. Passes for opx. May differ for cpx given broader P-T distribution in cpx training data.

Ship decision if passed. Resampling is ablated: move resampled joblibs to `models/ablation/<pipeline>/resampled/`, update per-pipeline `nb03_per_family_winners.json` with `ablation_tested: [resampled]` metadata.

Ship decision if failed. Resampling unexpectedly helped. Reconsider canonical status in favor of resampled models. Low probability for opx; moderate probability for cpx.

T04: N_AUG=1 beats N_AUG=5 on test RMSE

Hypothesis. Disabling Agreda-Lopez EPMA noise augmentation (N_AUG=1) gives lower test T and P RMSE than N_AUG=5 (the v6 / v7 default).

Design. Fit with N_AUG=1 and N_AUG=5 using same Optuna params and same split. Compare test RMSE for both targets, both tracks. Single-seed is sufficient for a sensitivity check.

Accept criterion. N_AUG=1 test RMSE <= N_AUG=5 test RMSE for both targets, both tracks. Inequality can be small.

Expected result (v9 sensitivity test for opx). N_AUG=1 wins. Passes.

Ship decision if passed. N_AUG=1 is canonical for that pipeline. `augment_dataframe` still exists in `src/features.py` for future users but is not called in the canonical path.

Ship decision if failed. Revisit N_AUG default for that pipeline. Would require re-running with augmentation enabled.

T05: Feature set winners stable

Hypothesis. The winning feature set per (model, target, track) combo is stable between v9 opx and v10 opx Optuna runs, OR between v10 cpx and v10 twopx Optuna runs (each pipeline tested independently).

Design. For opx: load `nb03_optuna_best_params.json` from v9. Re-fit each combo at the frozen params with 8-model roster. Compute multi-seed test RMSE. Identify winning feature set per combo. Compare to v9 winners.

For cpx, twopx, universal: run Optuna from scratch (~2.5 h each), then test whether raw/alr/pwlr winners are stable across re-runs with different seeds (compare seed 42 Optuna vs seed 43 Optuna).

Accept criterion. Winning feature set matches in >= 10 of 12 combos (allow 2 ties where means are within 1 std).

Expected result. Should match exactly for opx if Optuna seed and data are unchanged. Cpx/twopx/universal expected to show similar stability.

Ship decision if passed. v9 Optuna params are canonical for v10 opx. No re-optimization needed for opx. For cpx/twopx/universal, frozen params ship.

Ship decision if failed. Investigate data or Optuna version drift. Re-run Optuna for the unstable pipeline.

T06: Mineral-only ML is worse than paired-phase ML (opx/cpx pipelines only)

Hypothesis. opx-only ML has negative R^2 on ArcPL; cpx-only ML may be negative or barely positive. Confirms mineral-only models cannot be primary.

Design. Run opx-only / cpx-only ML predictions on ArcPL Kd-eq subset. Compare R^2 to Putirka 2-px eq36/39 and to their liquid-paired counterparts.

Accept criterion. Mineral-only R^2 < 0.3 AND paired-phase R^2 > mineral-only R^2.

Expected result (v9 ArcPL metrics for opx). opx-only stacked T R^2 = -0.567, P R^2 = -0.244. Passes.

Ship decision if passed. Mineral-only relegated to supplementary. Manuscript primary results use liq-paired or two-pyroxene tracks. Framing: "mineral-only is for cases where liquid composition is unavailable (phenocrysts, xenoliths) and is fundamentally limited."

Ship decision if failed. Unexpectedly useful mineral-only; promote to primary. Rewrite framing.

Not applicable to twopx or universal pipelines.

T07: Composition-conditional bias correction improves ArcPL T

Hypothesis. A ridge regression of ArcPL T residual on liq_Na2O, H2O_Liq, liq_TiO2, liq_K2O, Mg_num fitted by in-domain GroupKFold (not on ArcPL) reduces ArcPL T RMSE by >=5 C.

Design.

1. Compute in-domain train-set residuals via 5-fold GroupKFold CV on training data.
2. Fit `ridge_corr = RidgeCV().fit(X_composition, residuals)` where X_composition is the 5 features above, standardized.
3. Apply correction to ArcPL predictions: `T_corrected = T_predicted - ridge_corr.predict(X_arcpl_composition)`.
4. Report ArcPL T RMSE before and after correction.
5. Bootstrap 95% CI on the RMSE delta.

Accept criterion. ArcPL T RMSE decreases by >= 5 C with 95% CI excluding zero, using boosted as the base model.

Expected result (probe7 forest numbers for opx). Linear composition correction: 17.7% in-domain improvement, significant. Should translate to ~10 C ArcPL improvement on boosted (bigger absolute bias, similar fractional).

Ship decision if passed. Correction is canonical in NB07. Manuscript reports both raw and corrected numbers per probe7 recommendation. Correction is flagged as "in-domain trained, applied out-of-domain" — a reviewer-transparent design.

Ship decision if failed. Do not correct. Report raw +20.95 C ArcPL T bias with caveat about distribution shift and composition sensitivity. Flag liq_Na2O / H2O as OOD indicators in NB10.

Note: this test lives in NB07, not NB03. Decision logged to the per-pipeline NB03 test log for single-source-of-truth purposes.

T08: P piecewise bias correction improves test P

Hypothesis. The nb07 piecewise correction reduces test P RMSE beyond sampling noise (replicating v9 finding).

Design. Re-run nb07 piecewise correction on test set. Bootstrap 95% CI on RMSE delta.

Accept criterion. Test P RMSE decreases with 95% CI excluding zero.

Expected result (v9 `nb07_bias_correction_null_result.csv`). Delta = -0.27 kbar, CI excludes zero. Passes for opx.

Ship decision if passed. P piecewise correction ships in the canonical P inference pipeline. Manuscript Methods describes both the raw and corrected P. Correct the handoff claim of "NULL on both targets" — P is SIGNIFICANT.

Ship decision if failed. Unexpected deviation from v9. Investigate data shift.

Note: this test lives in NB07, not NB03.

T09: Fix stacking distributional mismatch (CV-predict at test time)

Hypothesis. Using `cross_val_predict(cv=5)` to generate base predictions at test time (not just training time) gives lower test T RMSE than using full-model predictions, by matching the distributional properties the meta model was trained on.

Design.

1. Legacy stacking: meta fit on OOF, test-time base predictions from full-data-fit base models. Record test RMSE.
2. Revised stacking: meta fit on OOF, test-time base predictions from `cross_val_predict(n_splits=5)`. Record test RMSE.
3. Compare.

Accept criterion. Revised test T RMSE < legacy test T RMSE by >= 1 C, OR revised test T RMSE within 1 C of legacy and revised ArcPL T RMSE no worse than legacy.

Expected result. Plausible small improvement on test; unknown effect on ArcPL. Low-confidence test.

Ship decision if passed. Revised stacking is canonical. Update `src/stacking.py` with `predict_with_cv` function. Update `docs/stacking_strategy.md` with the fix.

Ship decision if failed. Legacy stacking stays. Document the distributional mismatch in stacking_strategy.md as an acknowledged limitation.

Note: this test lives in NB04 (benchmark), decision logged to per-pipeline NB03 log.

T10: IsolationForest OOD flag correlates with residual magnitude

Hypothesis. Samples flagged OOD by IsolationForest have systematically larger absolute residuals than in-distribution samples.

Design. Compute IsolationForest OOD score on ArcPL n=197. Split ArcPL into 5 quintile bins by OOD score. Compute mean absolute T residual per bin. Test for monotonic trend via Spearman rank correlation.

Accept criterion. Spearman r >= 0.2 with p < 0.05 between bin rank and mean |T residual|.

Expected result (v9 `nb10_ood_isoforest.csv` for opx). OOD subset (n=24) has RMSE 61.9 vs in-domain (n=370) RMSE 64.6 — OOD is slightly BETTER. Unusual result; v9 data may not support this hypothesis.

Ship decision if passed. OOD flag is canonical for user warnings. Include in predictions CSV output.

Ship decision if failed. OOD flag is an informational diagnostic, not a reliability indicator. Document honestly in manuscript: "IsolationForest flags compositional outliers but does not strongly predict residual magnitude in our test. Report for transparency."

Note: this test lives in NB10, decision logged to per-pipeline NB03 log.

T11: New boosted/tree models (CatBoost, LightGBM) beat the v9 4-model roster

Hypothesis. CatBoost or LightGBM beats the best of (RF, ERT, XGB, GB) on pipeline-specific test T RMSE by at least 1% with 95% bootstrap CI excluding zero.

Design. Fit CatBoost and LightGBM under the same Optuna protocol (50 trials, TPE, seed 42). Compare multi-seed test RMSE (20 seeds) to best of v9 4-model roster.

Accept criterion. Either CatBoost or LightGBM achieves test T RMSE <= best_of_v9_4_models * 0.99, with 95% bootstrap CI on the delta excluding zero. Pipeline-specific.

Expected result. Likely to pass on at least one pipeline. CatBoost often wins on imbalanced tabular regression; LightGBM is faster but may not beat XGB.

Ship decision if passed. The winning model joins the canonical 8-model roster. If both pass, both ship.

Ship decision if failed. The new model is ablated for that pipeline but still trained and evaluated (kept in the 8-base ensemble shootout for T02). If both fail across all pipelines, consider dropping from pool entirely.

T12: Non-tree models (ElasticNet, MLP) contribute ensemble diversity

Hypothesis. ElasticNet and MLP contribute to ensemble performance even if they lose head-to-head to trees, because they provide prediction diversity for the stacked meta-model. Specifically: removing either from the 8-base pool increases ensemble test T RMSE.

Design. Train all 8 bases. Refit Ridge stacking over: (a) all 8, (b) 7 excluding ElasticNet, (c) 7 excluding MLP. Compare test T RMSE across the three ensembles.

Accept criterion. Ensemble (a) test T RMSE < both (b) and (c), with 95% bootstrap CI excluding zero on at least one comparison.

Expected result. Uncertain. ElasticNet/MLP may be too correlated with trees to add diversity, OR may provide the heterogeneity the v9 4-tree Ridge was missing. T12 tests this empirically.

Ship decision if passed. Non-tree bases ship in the canonical ensemble. Linear-SHAP on the Ridge meta (per NB06) reports which bases contribute.

Ship decision if failed. Non-tree bases are evaluated and reported for transparency but excluded from canonical ensemble. Document honestly: "ElasticNet and MLP were trained and evaluated but did not improve ensemble performance over the 6-tree or 4-tree configurations."

Test ordering

Per pipeline: run T01 through T12 in numeric order. Each test can fail without blocking downstream tests (they are independent). T07, T08, T09, T10 have execution in NB04/NB07/NB10 but decision logging in per-pipeline NB03 test log.

All results land in `results/v10_nb03_test_log.csv` with schema:

```
pipeline, test_id, hypothesis, metric, observed, threshold, passed, shipped,
timestamp, commit_sha
```

What this protocol is NOT

- Not a substitute for the existing multi-seed evaluation (20 seeds still computed for variance quantification)
- Not a substitute for LOSO / Cluster-KFold / TargetBin / LeaveOneRegionOut generalization (NB05 still runs all four)
- Not a gate on manuscript submission — failed tests become ablations, not show-stoppers, unless T01 fails (would require rethinking primary model)

Reviewer-facing value

By the time the manuscript reaches peer review, every pre-registered test has a logged outcome per pipeline. A reviewer who asks "why did you ship stacking but not resampling for opx but the opposite for cpx?" gets the answer: T02 passed for opx on its pre-registered criterion; T03 failed. For cpx the opposite happened. The criteria were set before data inspection, logged in this doc, committed to git before Phase C ran.

This is defensive, but it is the correct discipline for ML-in-petrology where the field has a history of post-hoc cherry-picking.

---

9. NB03 structure per pipeline

Each of the four NB03 variants follows this structure:

```
# Phase 3.0: Setup and imports
# Phase 3.1: Data loading
# Phase 3.2: Feature engineering (per feature set)
# Phase 3.3: Optuna hyperparameter search (T05 preregistered test)
# Phase 3.4: Multi-seed final training (20 seeds)
# Phase 3.5: Canonical winner selection
# Phase 3.6: Test T01 (primary model)
# Phase 3.7: Test T04 (N_AUG)
# Phase 3.8: Test T05 (feature stability)
# Phase 3.9: Test T03 (resampling ablation)
# Phase 3.10: Test T02 (ensemble methods)
# Phase 3.11: Test T06 (mineral-only supplementary; opx/cpx only)
# Phase 3.12: Tests T11, T12 (new model families)
# Phase 3.13: Serialize canonical models
# Phase 3.14: Write per_family_winners.json and manifests
```

Phase numbers harmonize with v9 structure; new tests add new phases rather than restructuring.

---

10. Cross-pipeline tests (separate)

Tests T07, T08, T09, T10 involve ArcPL / test-set inference per pipeline. They live in NB07 (bias), NB10 (OOD), NB04 (benchmark), not in NB03. But their pass/fail decisions are logged with NB03 test log per pipeline.

This keeps NB03 focused on training-time decisions and NB04/NB07/NB10 focused on inference-time decisions, while maintaining one test log per pipeline.

---

11. Universal-only tests T13 and T14

The universal masking model has two additional tests that do not apply to the opx/cpx/twopx pipelines because they depend on the masking architecture. Both are fully specified in `docs/universal_model_exploration.md`:

Test	Name	Hypothesis	Pass condition
T13	Graceful degradation	Universal R^2 scales monotonically with number of phases present (1 phase < 2 phases < 3 phases)	All three degradation tiers show RMSE ordering in expected direction; no catastrophic drop when phases are masked
T14	All-phase outperforms subset	A sample with opx+cpx+liq all filled gets lower RMSE from universal than the best specialized opx-liq or cpx-liq or twopx model on the same sample	Universal wins on >=60% of all-phase ArcPL samples by RMSE delta > 5 C (T) or > 0.3 kbar (P)

Both tests run ONLY in `nb03_universal_exploration.ipynb`. Neither test runs in opx, cpx, or twopx pipelines. Results logged to `results/universal/v10_nb03_test_log.csv` (separate from the primary test log).

Decision tree:

- If T13 fails: universal architecture is ablated and the exploration ends.
- If T13 passes but T14 fails: universal model is kept as an exploration artifact with honest negative framing ("tested but specialized models remain superior").
- If both pass: universal model becomes the lead novelty claim in the cpx paper.

See `docs/universal_model_exploration.md` Sections 5-7 for full rationale, evaluation protocol, and decision tree details.

---

12. T15: pre-registered regime-stratified pass condition (opx-liq pressure)

Scope: opx pipeline only (T15 for primary pipelines; T13 and T14 remain reserved for the universal exploration track — see Section 11).

Hypothesis: on the pre-registered four-bin P partition (edges `[0, 5, 15, 30, 100]` kbar, registered 2026-04-17 in `docs/preregistration/p_regime_preregistration.md`), at least one regime with n ≥ 20 shows a directional v10-outperforms-Putirka claim for the `P_kbar` target at the **two-axis honesty bar** (see v2 below).

Rationale: the opx paper's pre-registered headline claim is that the v10 thermobarometer is better calibrated in at least one geologically meaningful P regime, not globally. Section G.1c-extended produces the all-models regime CSV; the opx-liq subset of that CSV, filtered to the pre-registered bins, drives this test.

Pass condition (v2, revised 2026-04-18 in Chunk C): `results/v10_opx_per_regime_claims_audit_robust.csv` contains at least one row matching `target == 'P_kbar'` AND `n >= 20` AND `axis1_nonoverlap == True` AND `axis2_nonoverlap == True` (equivalently `robust_outperforms == True`). Both axes must pass: axis 1 (test-set bootstrap CI) quantifies sampling noise; axis 2 (20-seed RMSE spread) quantifies model-fit stochasticity. If both pass, the audit row counts as a "pre-registered headline claim passes, robustly." Otherwise the test fails and the manuscript's headline claim is reframed as calibration-domain characterization only (no directional claim).

Pass condition (v1, deprecated): the original single-axis pass condition read `results/v10_opx_per_regime_claims_audit.csv` and required only non-overlapping bootstrap 95% CI on the residual axis. Chunk C (2026-04-18) found that for non-deterministic cells (MLP/raw, CatBoost/raw) a single-seed point estimate could fall on the favorable tail of the across-seed RMSE distribution, inflating apparent performance. The robust audit adds the seed-axis check; v2 is the canonical pass condition for all future manuscript claims.

Source of truth: `results/v10_opx_per_regime_claims_audit_robust.csv` (produced by `scripts/v10_phase_g_chunkC_robust_audit.py`, which consumes `results/v10_opx_per_regime_benchmark.csv` from nb04 and `results/v10_chunkC_perseed_regime_rmse.csv` from `scripts/v10_phase_g_chunkC_seed_regime_probe.py`). This test does NOT re-run any training; it is a pure consistency check between the headline claim and the robust audit table.

Log format: results appended to `results/v10_nb03_test_log.csv` with `test_id=T15`, `pipeline=opx`, `target=P_kbar`, `track=opx_liq`. The `details` JSON field carries the winning regime, its v10 and Putirka RMSEs with CIs, seed-axis CI, and n.

Execution: `scripts/v10_nb03_test_t15.py` (see Appendix). One-shot script; no side effects beyond appending one row to the test log. After Chunk C the script reads the robust audit CSV.

---

13. T16-T18: Phase G.7 bias-correction tests

Numbering note. Original task-list terminology referenced "T13-T15" for the new bias-correction tests, but those IDs were already allocated (T13 and T14 to the universal exploration track in Section 11; T15 to the regime-stratified headline claim in Section 12). The Phase G.7 bias-correction work therefore registers as **T16, T17, T18**.

Scope: opx and cpx pipelines. Bias-correction is run on the eight aggregate-best cells of the multiseed summary (opx-liq T and P, opx-only T and P, cpx-liq T and P, cpx-only T and P). Universal and two-pyroxene pipelines are excluded (universal: scope blocked by T13/T14 decision gate; twopx: deferred to future work).

T16: At least one cell ships a bias correction

Hypothesis: on the 8 aggregate-best cells, at least one cell's canonical-seed (seed=42) run selects a winning form (A or B) that passes the ship criterion (`overall Delta-RMSE > SHIP_TOL` AND `max per-regime degradation <= SHIP_TOL`).

Pass condition: `results/v10_bias_correction_shipped.csv` has at least one row with `winner in {'A', 'B'}`.

Rationale: if no cell ships a correction, the manuscript reports bias-correction as null-result and recommends against deployment. A passing T16 unlocks the "applied the correction and confirmed it helps" headline.

T17: Shipped corrections are stable across seeds (majority rule)

Hypothesis: for every cell whose canonical-seed winner is not 'none', the same form ships on a **majority** (>10 of 20) of `SPLIT_SEEDS`. This guards against correction decisions that are artifacts of one favourable seed.

Pass condition: for each `(track, target)` with canonical winner `w in {'A', 'B'}`, `results/v10_bias_correction_per_seed.csv` has `n_seeds_ship_w = count(form == w AND regime == 'ALL' AND ships == True) > 10`. All cells that pass T16 must also pass T17 (joint).

Log format: one row per (pipeline, track, target) with `test_id=T17`, `value = n_seeds_ship`, `threshold = 10`.

T18: Form A and Form B parameter stability

Hypothesis: neither form's fitted parameters are artefacts of one particular fold split or bin-edge choice.

Pass condition (joint):

1. Form A edge sensitivity: `results/v10_bias_correction_edge_sensitivity.csv` shows for every cell the **maximum |Delta-RMSE swing|** across the three bin-edge sets `(base, inner_m1, inner_p1)` is < 0.5 kbar (P targets) or < 5 degrees C (T targets).
2. Form B stability: `results/v10_bias_correction_form_b_stability.csv` shows for every cell whose winner is 'B' (or where Form B fits successfully) the CV-reseed standard deviation of `alpha_L` and `alpha_R` is < 0.1 (quantile units).

Rationale: the manuscript must not recommend a correction that relies on a coincidental fold/edge choice. Thresholds are conservative by design.

Source of truth:

- T16, T17: `scripts/v10_phase_g_bias_correction_comprehensive.py` aggregation outputs.
- T18: `scripts/v10_phase_g_edge_sensitivity.py` + `scripts/v10_phase_g_form_b_stability.py` outputs.

Execution: tests are evaluated once the Phase G.7 D2/A4/A5 scripts have run end-to-end. Decision logic is recorded in `results/v10_nb03_test_log.csv` via `scripts/v10_nb03_test_t16_t18.py`.

T19: TabPFN v2 supplementary baseline smoke test

Hypothesis: the TabPFN v2 package (pinned `tabpfn>=2.0,<2.5`) is installed in `.venv-tabpfn` and can fit + predict on a 50-row opx_liq slice in under 30 seconds on CPU.

Pass condition (joint):

1. `tabpfn.__version__` reports `2.x` with `x < 5`, verified inside `scripts/tabpfn/opx_tb_nb03_test_tabpfn_smoke.py` before fitting.
2. `TabPFNRegressor` constructed via `create_default_for_version(ModelVersion.V2, ...)` or (fallback in tabpfn 2.0-2.4) `TabPFNRegressor(model_path='auto', ...)`, with the construction path logged.
3. Fit + predict on 50 opx_liq samples returns finite predictions with RMSE < 1000 kbar (sanity bound, not a performance threshold).

Rationale: the TabPFN baseline is a post-hoc supplementary benchmark, not a primary model. The smoke test is a tripwire: if it fails, the full `scripts/tabpfn/opx_tb_nb03_tabpfn_baseline.py` run does not start, so we cannot accidentally report numbers from the wrong model version (v2.5 non-commercial license + unrelated arxiv citation).

Source of truth:

- Construction + version check: `scripts/tabpfn/opx_tb_nb03_test_tabpfn_smoke.py`.
- Package pin: `requirements-tabpfn.txt`.

Execution: run `.venv-tabpfn/Scripts/python.exe scripts/tabpfn/opx_tb_nb03_test_tabpfn_smoke.py` before the main baseline run. Non-zero exit code blocks `scripts/tabpfn/opx_tb_nb03_tabpfn_baseline.py`.